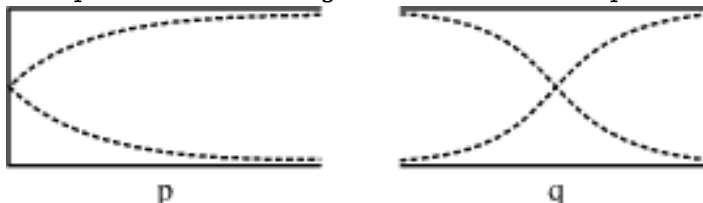
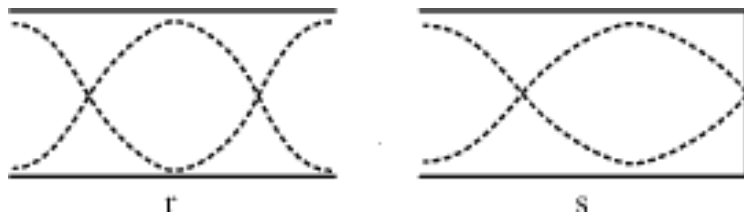


PHYSICS

1) The vibrations of four air columns are represented in the figure. The ratio of frequencies $n_p : n_q :$



$n_r : n_s$ is :- (Length in all case is same)



- (1) 12 : 6 : 3 : 4
- (2) 1 : 2 : 4 : 3
- (3) 4 : 2 : 3 : 1
- (4) 6 : 2 : 3 : 4

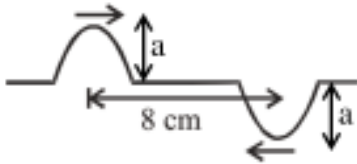
2) A tuning fork vibrating with a sonometer having 20 cm wire produces 5 beats per sec. The beat frequency does not change if the length of the wire is changed to 21 cm. The frequency of the tuning fork must be (Consider sonometer in fundamental mode of vibration) :-

- (1) 200 Hz
- (2) 210 Hz
- (3) 205 Hz
- (4) 215 Hz

3) If two tuning forks A and B are sounded together, they produce 4 beats per second. A is then slightly loaded with wax, they produce 2 beats per second when sounded again. The frequency of A is 256 Hz. The frequency of B will be :-

- (1) 250 Hz
- (2) 252 Hz
- (3) 260 Hz
- (4) 262 Hz

4) Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in figure. The speed of each pulse is 2 cm/s. After 2s, the total energy of the pulses will be :



- (1) zero
- (2) purely kinetic
- (3) purely potential
- (4) partly kinetic and partly potential

5) Two sources of sound placed close to each other, are emitting progressive waves given by $y_1 = 3 \sin 500\pi t$ and $y_2 = 6 \sin 508\pi t$

An observer located near these two sources will hear :-

- (1) 8 beats per second with intensity ratio 9 : 1 between waxing and waning.
- (2) 4 beats per second with intensity ratio 9 : 1 between waxing and waning.
- (3) 4 beats per second with intensity ratio 4 : 1 between waxing and waning.
- (4) 8 beats per second with intensity ratio 4 : 1 between waxing and waning.

6) Two tuning forks have frequencies 380 Hz and 384 Hz respectively. when they are sounded together, they produce 4 beats per second. After hearing the minimum sound, how long will it take to hear the maximum sound.

- (1) $\left(\frac{1}{2}\right)$ sec
- (2) $\left(\frac{1}{4}\right)$ sec
- (3) $\left(\frac{1}{8}\right)$ sec
- (4) $\left(\frac{1}{16}\right)$ sec

7) A string vibrates according to the equation $y = 5 \sin\left(\frac{2\pi x}{3}\right) \cos 20\pi t$, where x and y are in cm and t in sec. The distance between two adjacent nodes is :-

- (1) 3 cm
- (2) 4.5 cm
- (3) 6 cm
- (4) 1.5 cm

8) A wire of length ' ℓ ' having tension T and radius 'r' vibrates with fundamental frequency 'f'. Another wire of the same metal with length 2 ℓ having tension 2T and radius 2r will vibrate with fundamental frequency :-

- (1) f

(2) $2f$

(3) $\frac{f}{2\sqrt{2}}$

(4) $\frac{f}{2}\sqrt{2}$

9) A wave is represented by the equation

$$y = 10\sin 2\pi(100t - 0.02x) + 10\sin 2\pi(100t + 0.02x).$$

The maximum amplitude and loop length are respectively :-

(1) 20 units and 30 units

(2) 20 units and 25 units

(3) 30 units and 20 units

(4) 25 units and 20 units

10) Third overtone of a closed organ pipe is in unison with fourth harmonic of an open organ pipe. Find the ratio of lengths of the pipes.

(1) $\frac{7}{8}$

(2) $\frac{8}{7}$

(3) $\frac{7}{10}$

(4) $\frac{10}{7}$

11) A tuning fork of frequency 500 Hz is vibrated just above the tube of 80 cm length. ($V_{\text{sound}} = 300$ m/s)

(P)	Max. number of resonance possible	(A)	3
(Q)	Minimum water level when resonance occurs	(B)	65 cm
(R)	Maximum water level when resonance occurs.	(C)	5 cm
		(D)	2

(1) P - A, Q - D, R - B

(2) P - D, Q - B, R - C

(3) P - A, Q - C, R - B

(4) P - D, Q - C, R - B

12) If a pipe gives notes of frequencies 255, 425 and 595, what is fundamental frequency of the pipe and its type?

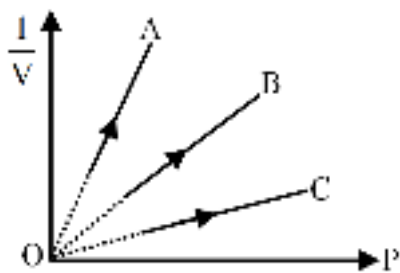
(1) 85, open pipe

(2) 17, closed pipe

(3) 51, open pipe

(4) 85, closed pipe

13) Figure shows the isotherms of a fixed mass of an ideal gas at three temperatures T_A , T_B and



T_C then

- (1) $T_A > T_B > T_C$
- (2) $T_A < T_B < T_C$
- (3) $T_B < T_A < T_C$
- (4) $T_A = T_B = T_C$

14) Two spherical vessel of equal volume, are connected by a narrow tube. The apparatus contains an ideal gas at one atmosphere and 300 K. Now if one vessel is immersed in a bath of constant temperature 600K and the other in a bath of constant temperature 300K. Then the common pressure



will be :-

- (1) 1 atm
- (2) $\frac{4}{5}$ atm
- (3) $\frac{4}{3}$ atm
- (4) $\frac{3}{4}$ atm

15) A gas has molar heat capacity $C = 4.5 R$ in the process $PT = \text{constant}$. The number of degrees of freedom of molecules in the gas is :-

- (1) 4
- (2) 3
- (3) 6
- (4) 5

16) **Assertion (A) :-** Blue star is at high temperature than red star.

Reason (R) :- Wien's displacement law states that $T \propto (1/\lambda_m)$.

- (1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
- (2) **(A)** is correct but **(R)** is not correct
- (3) **(A)** is incorrect but **(R)** is correct
- (4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

17) A monoatomic ideal gas, initially at temperature T_1 is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If ℓ_1 and ℓ_2 are the length of gas column, before and after the expansion

respectively, then the value of $\frac{T_1}{T_2}$ will be :

(1) $\left(\frac{\ell_1}{\ell_2}\right)^{2/3}$

(2) $\left(\frac{\ell_2}{\ell_1}\right)^{2/3}$

(3) $\frac{\ell_2}{\ell_1}$

(4) $\frac{\ell_1}{\ell_2}$

18) For an isothermal expansion of a perfect gas, the value $\frac{\Delta P}{P}$ is equal to

(1) $-\gamma^{1/2} \frac{\Delta V}{V}$

(2) $-\frac{\Delta V}{V}$

(3) $-\gamma \frac{\Delta V}{V}$

(4) $-\gamma^2 \frac{\Delta V}{V}$

19) A ball of mass 0.2 kg is thrown vertically upwards by applying a force by hand. If the hand moves 0.2 m while applying the force and the ball goes upto 2m height further, find the magnitude of the force. Consider $g = 10 \text{ m/s}^2$:

(1) 4 N

(2) 16 N

(3) 20 N

(4) 22 N

20) A block slides down an inclined plane of inclination θ with constant velocity. It is then projected up the inclined plane with an initial speed u . The distance up the plane it will move before coming to rest is

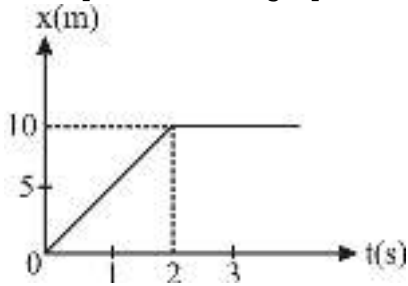
(1) $\frac{u^2}{g}$

(2) $\frac{u^2}{g \sin \theta}$

(3) $\frac{u^2}{2g \sin \theta}$

(4) $\frac{u^2}{4g \sin \theta}$

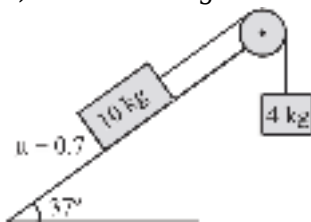
21) The position-time graph of a body of mass 2 kg is shown in :



What is the impulse on the body at $t = 2$ sec. ?

- (1) 5 kg ms^{-1}
- (2) $+ 10 \text{ kg ms}^{-1}$
- (3) -10 kg ms^{-1}
- (4) zero

22) In the arrangement shown in the figure ($\sin 37^\circ = 3/5$) :-



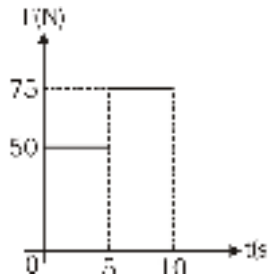
- (1) Direction of force of friction is up the plane
- (2) The magnitude of force of friction is zero
- (3) The tension in the string is 20 N
- (4) Magnitude of force of friction is 56 N.

23) A given object takes n times as much time to slide down a 45° rough incline as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is given by :-

- (1) $\left(1 - \frac{1}{n^2}\right)$
- (2) $\frac{1}{1 - n^2}$
- (3) $\sqrt{\left(1 - \frac{1}{n^2}\right)}$
- (4) $\sqrt{\frac{1}{1 - n^2}}$

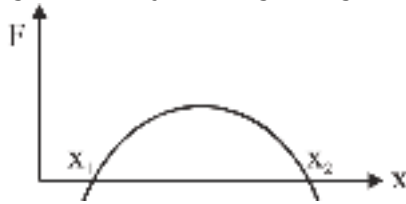
24)

A force represented as show in figure acts on a body having a mass of 16 kg. The velocity of the body at $t = 10$ s, if the body starts from rest :-



- (1) 100 m/s
- (2) 50 m/s
- (3) 49 m/s
- (4) 39 m/s

25) The force acting on a body moving along x-axis varies with the position of the particle as shown



in the figure.

The body is in stable equilibrium at :-

- (1) $x = x_1$
- (2) $x = x_2$
- (3) both x_1 and x_2
- (4) neither x_1 nor x_2

26) A position dependent force $F = 7 - 2x + 3x^2$ newton acts on a small body of mass 2 kg and displaces it from $x = 0$ to $x = 5$ m. The work done in joules is

- (1) 70
- (2) 270
- (3) 35
- (4) 135

27) A body of mass m is accelerated uniformly from rest to a speed v in a time T . The instantaneous power delivered to the body as a function of time t , is given by :

- (1) $\frac{mv^2}{T^2}t$
- (2) $\frac{mv^2}{T^2}t^2$
- (3) $\frac{1}{2} \frac{mv^2}{T^2}t^2$
- (4) $\frac{1}{2} \frac{mv^2}{T^2}t$

28) A object of mass 2 kg is released from height 10 m. If the object experience air drag with drag force of 10 N acting in opposite direction of motion. Final velocity-

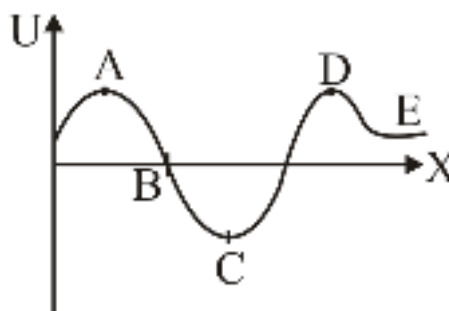
- (1) 12 m/s
- (2) 10 m/s
- (3) 18 m/s
- (4) 20 m/s

29) A vehicle of mass m whose engine delivers a constant power P , has an acceleration $a = \frac{P}{mv}$. In order to increase velocity from v_1 to v_2 , distance travelled will be :

- (1) $\frac{3P}{m}(v_2^2 - v_1^2)$
- (2) $\frac{m}{3P}(v_2^3 - v_1^3)$
- (3) $\frac{m}{3P}(v_2^2 - v_1^2)$
- (4) $\frac{m}{3P}(v_2 - v_1)^3$

30) A body of mass M is dropped from a height h on a sand floor. If the body penetrates x cm into the sand, the average resistance offered by the sand to the body is :

- (1) $Mg \left(\frac{h}{x} \right)$
- (2) $Mg \left(1 + \frac{h}{x} \right)$
- (3) $Mgh + Mgx$
- (4) $Mg \left(1 - \frac{h}{x} \right)$



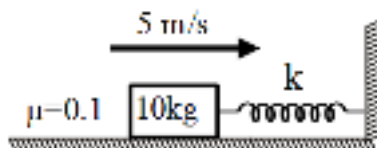
31) Match the following column for suitable equilibrium.

Column-I		Column-II	
(A)	At point A	(P)	Stable equilibrium
(B)	At point E	(Q)	Unstable equilibrium
(C)	At point C	(R)	Neutral equilibrium
(D)	At point D	(S)	None

- (1) A-Q, B-R, C-Q, D-P
- (2) A-P, B-Q, C-R, D-S
- (3) A-Q, B-R, C-P, D-Q

(4) A-S, B-R, C-Q, D-P

32) A block of mass 10 kg is projected horizontally on a rough horizontal floor. The coefficient of friction between the block and the floor is 0.1. The block strikes a light spring of stiffness $k = 100$ N/m with a velocity 5 m/s. The maximum compression of the spring is (in approximation)



- (1) 1.5 m
- (2) 2.5 m
- (3) 3.5
- (4) 4.5 m

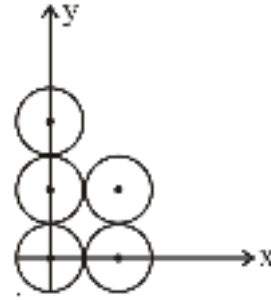
33) A small block of mass m is kept on a rough inclined surface of inclination θ fixed in a elevator. The elevator goes down with a uniform velocity v and the block does not slide on the wedge. The work done by the force of friction on the block in time t will be :-

- (1) zero
- (2) $-mgvt \cos^2\theta$
- (3) $-mgvt \sin^2\theta$
- (4) $mgvt \sin^2\theta$

34) Rate of doing work by force acting on a particle which starts its motion from origin along x-axis depends on a position x of particle and this rate is equal to $2x$. Expression of velocity of particle at any position x is :-

- (1) $\left(\frac{3x^2}{m}\right)^{1/3}$
- (2) $\left(\frac{3x^2}{2m}\right)^{1/3}$
- (3) $\left(\frac{2mx}{9}\right)^{1/2}$
- (4) $\left(\frac{mx^2}{3}\right)^{1/2}$

35) Five uniform circular plates, each of diameter b and mass m are laid out in a pattern shown the



y-coordinate of centre of mass of the five plate system will be :- Origin

- (1) $\frac{b}{5}$
- (2) $\frac{2b}{5}$
- (3) $\frac{4b}{5}$
- (4) $\frac{3b}{5}$

36) A projectile of mass 6 kg is projected from the ground with velocity 40 m/s at 45° with the horizontal. At the highest point, it explodes into two pieces. One of pieces has mass 4 kg and the other mass is 2kg. Both the pieces fly off after explosion. Mass 4 kg falls at a distance of 200 m from the point of projection. Find the distance of 2 kg from the point of projection where it strike the ground ($g = 10 \text{ m/s}^2$)

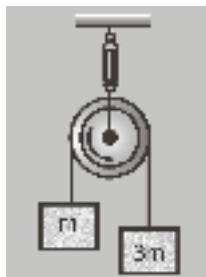
- (1) 70 m
- (2) 80 m
- (3) 90 m
- (4) 160 m

37) A uniform rod of mass M & length L is bent in to a quarter of a circle. Find the distance of Centre of mass of the shape from it's Centre?

- (1) $\frac{2\sqrt{3}L}{\pi^2}$
- (2) $\frac{4\sqrt{2}L}{\pi^2}$
- (3) $\frac{3\sqrt{2}L}{\pi^2}$
- (4) $\frac{4\sqrt{3}L}{\pi^2}$

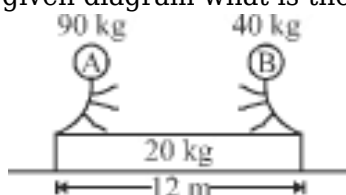
38)

If the system is released, then the acceleration of the centre of mass of the system :-



- (1) $g/4$
- (2) $g/2$
- (3) g
- (4) $2g$

39) In the given diagram what is the distance travelled by the plank when A and B exchange their



positions:

- (1) 2 m
- (2) 4 m
- (3) 4.8 m
- (4) 5.4 m

40) A force acts on a 3 gm particle in such a way that the position of the particle as a function of time is given by $x = 3t - 4t^2 + t^3$, where x is meters and t is in seconds. The work done during the first 4 second is-

- (1) 384 mJ
- (2) 168 mJ
- (3) 528 mJ
- (4) 541 mJ

41) 1 joule of energy is to be converted into new system of units in which length is measured in 10 metre, mass in 10 kg and time in 1 minute. The numerical value of 1 J in the new system :

- (1) 36×10^{-4}
- (2) 36×10^{-3}
- (3) 36×10^{-2}
- (4) 36×10^{-1}

42) A wave is propagating along x-axis in a medium, so that displacement y of particle at any

position x , at any time t is given by : $y = \frac{a}{t^2} e^{-\left[\frac{t}{b} - \frac{x}{c}\right]}$, where a , b and c are constants.

Choose the correct option (s) :-

- (1) dimension of b is $[L]$

- (2) dimension of a is $[LT^2]$
 (3) dimension of c is $[T]$
 (4) $e^{-\left(\frac{t}{b} - \frac{x}{c}\right)}$ is not dimensionless

43) If P, Q, R are physical quantities, having different dimensions, which of the following combinations can never be meaningful quantity :-

- (a) $\frac{(P-Q)}{R}$ (b) $PQ - R$
 (c) $\frac{PQ}{R}$ (d) $\frac{(R+Q)}{P}$

- (1) (a) and (d)
 (2) (b) and (c)
 (3) (b) and (d)
 (4) (c) and (d)

44) The resistance is $R = \frac{V}{I}$ where $V = (100 \pm 5)$ volt and $i = (10 \pm 0.2)$ ampere. What is the total error in R ?

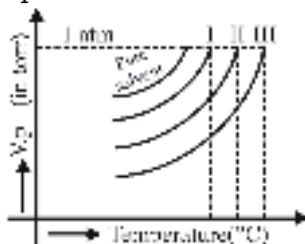
- (1) 5 %
 (2) 7 %
 (3) 5.2 %
 (4) $\left(\frac{5}{2}\right)\%$

45) In a vernier callipers, 10 VSD coincides with 8 MSD, then the least count of vernier calliper is :
 [1 MSD = 1 mm]

- (1) 1×10^{-4} m
 (2) 2×10^{-4} m
 (3) 1×10^{-3} m
 (4) 8×10^{-4} m

CHEMISTRY

1) The vapour pressure curves of the same non-volatile, non-electrolyte solute in the same solvent are shown. The curves are parallel to each other and do not intersect. The concentrations of



solutions are in order of :

- (1) $I < II < III$

- (2) $I = II = III$
- (3) $I > II > III$
- (4) $I > III > II$

2) Mole fraction of the toluene in the vapour phase which is in equilibrium with a solution of benzene ($p^\circ = 120$ Torr) and toluene ($p^\circ = 80$ Torr) having 2.0 mol of each is :-

- (1) 0.50
- (2) 0.25
- (3) 0.60
- (4) 0.40

3) Calculate the mass of urea (NH_2CONH_2) required to prepare 2.06 kg of 0.5 molal aqueous solution?

- (1) 90 g
- (2) 120 g
- (3) 60 g
- (4) 45 g

4) The molal elevation constant of water = $0.52 \text{ K kg mol}^{-1}$. The boiling point of 1.0 molal aqueous KCl solution (assuming complete dissociation of KCl), should be :-

- (1) 100.52°C
- (2) 101.04°C
- (3) 99.48°C
- (4) 98.96°C

5) An aqueous solution freezes at -0.186°C ($K_f = 1.86^\circ$, $K_b = 0.512^\circ$). What is the elevation in boiling point :-

- (1) 0.186
- (2) 0.512
- (3) $\frac{0.512}{1.86}$
- (4) 0.0512

6) Which one is incorrect for ideal solutions

- (1) $\Delta S_{\text{mix.}} = 0$
- (2) $\Delta G_{\text{mix.}} = 0$
- (3) They form azeotropes
- (4) All are incorrect

7)

Consider the following cases :-

Case I : 2M CH_3COOH solution in benzene at 27°C where there is dimer formation to the extent of 100%.

Case II : 0.5 M KCl aqueous solution at 27°C , which ionises 100%

Which is correct ?

- (1) Both are isotonic
- (2) I is hypertonic
- (3) II is hypertonic
- (4) None is correct

8) Two solutions of KNO_3 and CH_3COOH are prepared separately. Molarity of both is 0.1M and osmotic pressures are P_1 and P_2 respectively. The correct relationship between the osmotic pressures is

- (1) $P_2 > P_1$
- (2) $P_1 = P_2$
- (3) $P_1 > P_2$
- (4) $\frac{P_1}{P_1 + P_2} = \frac{P_2}{P_1 + P_2}$

9) **Assertion :** Solubility of gas increases with increase in pressure.

Reason : Henry constant of gas increase with increase in pressure.

- (1) **Assertion** is correct and **Reason** is incorrect
- (2) **Assertion** is incorrect and **Reason** is correct
- (3) Both **Assertion** and **Reason** are correct but **Reason** is not correct explanation of **Assertion**.
- (4) **Assertion** and **Reason** both are correct and **Reason** is correct explanation of **Assertion**.

10) Predict which of the following mixtures will show a positive deviation from Raoult's law?

- (1) Methanol and acetone
- (2) Chloroform and acetone
- (3) Nitric acid and water
- (4) Phenol and aniline

11) **Statement-I :-** 0.1 m aqueous solution of glucose has same increament in boiling point as in 0.1 m aqueous solution of urea.

Statement-II :- K_b for both 0.1 m aqueous solution of glucose and 0.1 m aqueous solution of urea are different.

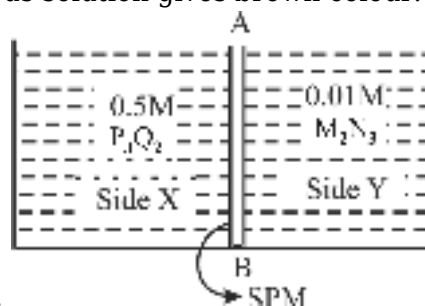
- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect.
- (3) **Statement I** is incorrect but **Statement II** is correct.
- (4) Both **Statement I** and **Statement II** are correct.

12) 17.4 % (mass / volume) K_2SO_4 solution at 27°C is isotonic to 5.85% (mass / volume) NaCl solution at 27°C . If NaCl is 100% ionised, what is % ionisation of K_2SO_4 in aqueous solution ?

[At wt. of K = 39, Na = 23] :-

- (1) 25%
- (2) 75%
- (3) 50%
- (4) None of these

13) P_3Q_2 ($i = 5$) When reacted with M_2N_3 ($i = 5$) in aqueous solution gives brown colour. These are



separated by a semipermeable membrane AB as shown. Due to osmosis there is :

- (1) brown colour formation in side X
- (2) brown colour formation in side Y
- (3) Formation in both of side X and Y
- (4) no brown colour formation.

14) Which of the following is correct regarding depression in freezing point?

- (a) Vapour pressure of solution is less than that of pure solvent
- (b) Vapour pressure of solution is more than that of pure solvent
- (c) Only solute molecules solidify at the freezing point.
- (d) Only solvent molecules solidify at the freezing point.

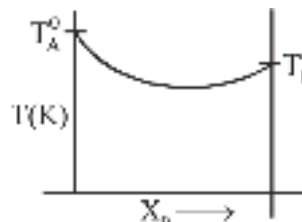
Correct are :-

- (1) a, d
- (2) c, a
- (3) b, d
- (4) b, c

15) 20 g of naphthoic acid ($C_{11}H_8O_2$) ($M_w = 172$ g/mol) is dissolved in 50 g of benzene then freezing point depression of 2K is observed for solution. The vant Hoff factor will be (K_f for benzene = 1.72 K.m $^{-1}$)

- (1) 0.75
- (2) 2
- (3) 0.5
- (4) 3

16) The diagram given below represents boiling point composition diagram of solution of



components A and B. Which is correct among the following ?

- (1) The solution shows negative deviation.
- (2) A-B interaction are stronger than A-A and B-B
- (3) The solution is ideal solution.
- (4) The solution show positive deviation.

17) Bonding in which of the following diatomic molecule become(s) stronger, on the basis of MO Theory, by removal of an electron ?

- (A) NO (B) N₂ (C) O₂
 (D) C₂ (E) B₂

Choose the most appropriate answer from the options given below :-

- (1) (A), (B), (C) only
- (2) (B), (C), (E) only
- (3) (A), (C) only
- (4) (D) only

18) The solubility of inert gases in water is due to

- (1) Keesom attraction
- (2) Debye attraction
- (3) London force
- (4) Ion-dipole attraction

19) How many statements is/are correct ?

- (i) Van der waal's forces are responsible for the formation of dry ice (molecular crystal).
- (ii) Ion-dipole intractions are responsible for hydration of ions.
- (iii) In solidified noble gas, the attraction between atoms arises entirely from weakest dispersion (London)forces.
- (iv) Boiling point of IC₄ is less than boiling point of Br₂.
- (v) Ionic bonds are directional in nature.

- (1) 4
- (2) 5
- (3) 3
- (4) 2

20) Which is not related to hydrogen bonding?

- (1) Boiling point of B(CH₃)₃ > BF₃
- (2) Zig-Zag structure of HF(s)

- (3) layered structure $\text{H}_3\text{BO}_{3(s)}$
- (4) Stability of chloral hydrate

21) Which of the following order is **CORRECT** for boiling point ?

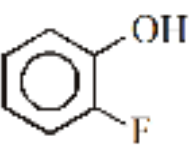
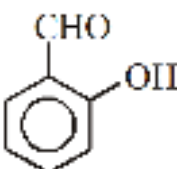
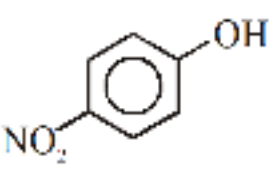
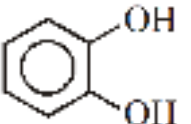
- (1) $\text{NH}_3 > \text{PH}_3$
- (2) $\text{CH}_3 - \text{O} - \text{CH}_3 > \text{C}_2\text{H}_5\text{OH}$
- (3) o-hydroxybenzaldehyde > p-hydroxybenzaldehyde
- (4) None of these

22) Choose the correct order for the given properties.

- (1) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: covalent character order.
- (2) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: melting point order
- (3) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: lattice energy order
- (4) $\text{NaF} > \text{MgF}_2 > \text{AlF}_3$: order of polarising power of cation.

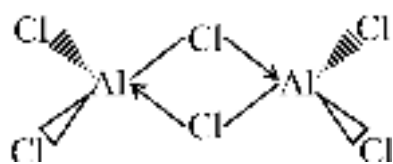
23)

Intramolecular hydrogen bond is not present in :

- (1) 
- (2) 
- (3) 
- (4) 

24)

Consider the structure of Al_2Cl_6 and choose correct statement



- (1) It is planar molecule
- (2) Al uses its sp^3 hybrid orbital to form bond with chlorine

- (3) It has $3c-2e^-$ and $2c-2e^-$ bonds
(4) Al_2Cl_6 is electron deficient molecule

25) In which of the following $p\pi - d\pi$ back bonding present ?

- (a) $N(SiH_3)_3$ (b) $P(SiH_3)_3$ (c) SCl_2 (d) $O(SiH_3)_2$

- (1) a, b
(2) b, d
(3) a, d
(4) b, c

26) Select correct information regarding $B_3N_3H_6$.

- (1) $B_3N_3H_6$ is non-planar compound
(2) $B_3N_3H_6$ is aromatic compound
(3) In $B_3N_3H_6$ back bonding is not available
(4) $B_3N_3H_6$ is less reactive than C_6H_6

27) Boron forms BX_3 type of halides. The CORRECT increasing order of Lewis-acid strength of these halides is

- (1) $BF_3 > BCl_3 > BBr_3 > BI_3$
(2) $BI_3 > BBr_3 > BCl_3 > BF_3$
(3) $BF_3 > BI_3 > BCl_3 > BBr_3$
(4) $BF_3 > BCl_3 > BI_3 > BBr_3$

28) Which of the following oxy acid can not act as an ortho acid :

- (1) H_3BO_3
(2) H_3PO_4
(3) H_2SO_4
(4) None of these

29) How many peroxy linkage are present in perchloric acid ?

- (1) 0
(2) 2
(3) 1
(4) 3

30) Select correct statement about hydrolysis of BCl_3 and NCl_3

- (1) NCl_3 is hydrolysed and gives $HOCl$ but BCl_3 is not hydrolysed.
(2) Both NCl_3 and BCl_3 on hydrolysis gives HCl
(3) NCl_3 on hydrolysis gives $HOCl$ but BCl_3 gives HCl

(4) Both NCl_3 and BCl_3 on hydrolysis gives HOCl

31) Which of the following statement(s) is/are **CORRECT**?

- (1) BiCl_5 is more stable than BiF_5
- (2) PbCl_2 is more stable than GeCl_2 and SnCl_2
- (3) PbI_4 exist but PbCl_4 does not
- (4) Tl^{+3} ion is more stable than Tl^{+1} ion

32) Which of the following statements is correct ?

- (1) BCl_3 and AlCl_3 are both Lewis acids and BCl_3 is stronger than AlCl_3
- (2) BCl_3 and AlCl_3 both Lewis acids and AlCl_3 is stronger than BCl_3
- (3) BCl_3 and AlCl_3 are both equally strong Lewis acids
- (4) Both BCl_3 and AlCl_3 are not Lewis acids.

33) Identify the **incorrect** statement :

- (1) Boron trioxide is acidic, aluminium and gallium trioxides are amphoteric while indium and thallium trioxides are basic.
- (2) Boiling point of Ga is maximum in 13th group
- (3) Two isotopes of Boron, namely ^{10}B (19%) and ^{11}B (81%) are known.
- (4) The relative stability of group 13 elements in +1 oxidation state varies as $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$.

34) Oxide of element A + H_2O -----> solution (X)

Oxide of element B + H_2O -----> solution (Y)

Solution 'X' turns blue litmus paper to red and solution 'Y' turns red litmus to blue then element A & B will be respectively -

- (1) K & C
- (2) Mg & S
- (3) Cl & Na
- (4) Ca & Na

35) The first and second ionisation enthalpy and first e^- gain enthalpy in kJ/mol of few elements are given below :

Element	IE_1	IE_2	ΔH_{eg}
M	419	3051	- 48
N	1681	3374	- 328
O	1008	1846	- 295
P	2372	5251	+ 48

The incorrect statement regarding these elements is -

- (1) Element 'M' is an alkali metal.

- (2) Element 'P' is a noble gas element.
- (3) Element 'N' is a highest reactive non metal in given element.
- (4) Element 'O' is an alkaline earth metal.

36) One mole of $F_{(g)}$ atoms are ionised to $F_{(g)}^-$ and energy released is X joule then -

- (1) X joule energy is sufficient to ionise 1 mole $Cl_{(g)}$ into $Cl_{(g)}^+$
- (2) X joule energy is sufficient to ionise 1 mole $F_{(g)}$ into $F_{(g)}^+$
- (3) X joule energy is sufficient to ionise 1 mole of $Cl_{(g)}$ into $Cl_{(g)}^+$ as well as 1 mole of $F_{(g)}$ into $F_{(g)}^+$
- (4) X joule energy is less than energy required to ionise 1 mole of $Cl_{(g)}$ into $Cl_{(g)}^+$ as well as 1 mole of $F_{(g)}$ into $F_{(g)}^+$

37) **Assertion (A)** : Across a period non-metallic character increases.

Reason (R) : Across a period electronegativity increases.

Choose the correct option among following -

- (1) Both (A) and (R) are correct but (R) is not correct explanation of (A).
- (2) (A) is correct but (R) is incorrect.
- (3) (A) is incorrect but (R) is correct.
- (4) Both (A) and (R) are correct but (R) is correct explanation of (A).

38) Consider the following processes :

- (i) $O_{(g)} \rightarrow O_{(g)}^-$ (ii) $F_{(g)} \rightarrow F_{(g)}^-$
- (iii) $Cl_{(g)} \rightarrow Cl_{(g)}^-$ (iv) $O_{(g)} \rightarrow O_{(g)}^{2-}$

The according to given in formation which statement is not correct ?

- (1) Process (iii) is more exothermic than process (i) and (ii).
- (2) Process (i) is less exothermic than process (ii)
- (3) Process (i), (ii) and (iii) are exothermic while process (iv) is endothermic
- (4) Process (i) and (iii) are exothermic while process (ii) and (iv) is endothermic

39) **Assertion (A)** : Electron gain enthalpy for N is +ve while for P is -ve.

Reason (R) : Smaller size of 2p orbital of N in which there is a considerable electronic repulsion while larger size of 3p orbital of P in which electronic repulsion is not considerable.

Choose the correct option among following -

- (1) Both (A) and (R) are correct but (R) is not correct explanation of (A).
- (2) (A) is correct but (R) is incorrect.
- (3) (A) is incorrect but (R) is correct.
- (4) Both (A) and (R) are correct but (R) is correct explanation of (A).

40) **Assertion (A)** : In halogen family electron affinity decreases from fluorine to Iodine.

Reason (R) : On increasing size nuclear attraction decreases in halogen family.

Choose the correct option among following -

- (1) Both (A) and (R) are correct but (R) is not correct explanation of (A).

- (2) (A) is correct but (R) is incorrect.
 (3) (A) is incorrect but (R) is correct.
 (4) Both (A) and (R) are correct but (R) is correct explanation of (A).

41) Suppose a gaseous mixture of F, Cl, Br and I is irradiated with photons of frequency appropriate to ionise Cl, then which ions(s) will be present in the mixture ?

- (1) F^+ only
 (2) F^+ , Cl^+ , Br^+ only
 (3) Cl^+ only
 (4) Cl^+ , Br^+ and I^+ only

42) Match the column-I and Column-II.

	Column-I (Electronic configuration)		Column-II (IE in kJ/mol)
(A)	$[Ne] 3s^2, 3p^1$	(P)	1000
(B)	$[Ne] 3s^2$	(Q)	737
(C)	$[Ne] 3s^2, 3p^3$	(R)	577
(D)	$[Ne] 3s^2, 3p^4$	(S)	1012

- (1) A-Q, B-S, C-P, D-R
 (2) A-R, B-Q, C-S, D-P
 (3) A-P, B-Q, C-R, D-S
 (4) A-Q, B-R, C-P, D-S

43) Identify the incorrect match.

	Name		IUPAC official name
(A)	Unnilibium	(i)	Nihonium (Nh)
(B)	Unnilpentium	(ii)	Nobelium (No)
(C)	Ununtrium	(iii)	Livermorium (Lv)
(D)	Ununhexium	(iv)	Bohrium (Bh)

- (1) A-ii
 (2) B-iv
 (3) D-iii
 (4) C-i

44) An element having e^- configuration

$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10} 4f^{14}, 5s^2 5p^6 5d^1, 6s^2$
 belongs to

- (1) s-block
- (2) p-block
- (3) d-block
- (4) f-block

45) Atomic number of an element 46. In the same group the atomic number of elements placed below and above given element will be -

- (1) 28, 64
- (2) 64, 28
- (3) 28, 78
- (4) 78, 28

BIOLOGY

1) Which one of the following option gives the correct categorisation of formed elements with respect to their count and function :-

	(A) Element	(B) Count	(C) Function
(1)	Erythrocyte	55 million/mm ³	Transport of gas
(2)	Leucocyte	6000 - 8000/mm ³	Clotting of Blood
(3)	Thrombocyte	1.5 - 3.5 lakh/mm ³	Immunity
(4)	Platelets	1.5 - 3.5 lakh/mm ³	Coagulation

- (1) 1
- (2) 2
- (3) 3
- (4) 4

2) Identify the correct match from the columns I, II and III.

Column-I		Column-II		Column-III	
(1)	First heart sound	(a)	DUB	(i)	Blood pumped per minute by each ventricle.
(2)	Cardiac output	(b)	5L	(ii)	Blood pumped by each ventricle in one contraction.
(3)	Second heart sound	(c)	70 ml	(iii)	Ventricular systole.
(4)	Stroke volume	(d)	LUB	(iv)	Ventricular diastole.

- (1) 1-d-i, 2-b-iii, 3-c-ii, 4-a-iv

(2) 1-d-iii, 2-b-i, 3-a-iv, 4-c-ii

(3) 1-d-iii, 2-b-i, 3-a-ii, 4-c-iv

(4) 1-a-ii, 2-b-iii, 3-c-i, 4-d-iv

3) Read the following properties of blood corpuscles:-

(a) They are colourless due to lack of haemoglobin

(b) They are non-nucleated

(c) They are relatively lesser in number which averages $6000-8000 \text{ mm}^{-3}$ of blood

(d) They have long life span approx 120 days.

Out of these which properties are not shown by leukocytes ?

(1) 'a' and 'b'

(2) 'b' and 'c'

(3) 'b' and 'd'

(4) 'c' and 'd'

4) Read the following statements carefully :

(i) Anti Rh-antibodies administered to mother immediately after the delivery of first child.

(ii) An injury or trauma stimulate leucocyte in blood to release factors to activate mechanism of coagulation.

(iii) Exchange of nutrients, gases etc. between blood and cells always occurs through lymph.

(iv) All reptiles have 3 chambered heart.

Find out correct statements :

(1) i, ii, iii

(2) ii, iii, iv

(3) i, iii

(4) ii, iv

5) Match the following column and select the correct option.

Column-I		Column-II	
(a)	Chordae Tendinae	(i)	Minute fibres throughout the ventricular musculature of the respective side
(b)	Sino-atrial node	(ii)	Bundle of fibres immediately divides into right and left bundle branch.
(c)	Bundle of His	(iii)	Fibres make AV valve unidirectional
(d)	Purkinje fibres	(iv)	Generate the maximum number of action potential

	(a)	(b)	(c)	(d)
(1)	(iii)	(iv)	(i)	(ii)
(2)	(ii)	(iii)	(i)	(iv)
(3)	(ii)	(iv)	(i)	(iii)

(4)	(iii)	(iv)	(ii)	(i)
-----	-------	------	------	-----

- (1) 1
- (2) 2
- (3) 3
- (4) 4

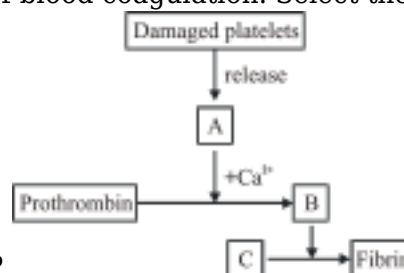
6) ECG is graphical representation of the electrical activity of the heart during a cardiac cycle. Read following statements.

- (i) To obtain a standard ECG, a patient is connected to the machine with three electrical leads one to each wrist and to the left ankle.
- (ii) P wave represents the repolarisation of the atria which leads to the contraction of both the atria.
- (iii) T wave represents the return of the ventricles from excited to normal state.
- (iv) Ventricular contraction starts shortly after Q wave.

Which of the following option contain correct statements.

- (1) (i), (ii), (iii), (iv)
- (2) (i), (ii), (iii)
- (3) (i) and (ii) only
- (4) (i), (iii), (iv)

7) The figure below shows steps of blood coagulation. Select the option giving correct identification



together with what it represents ?

- (1) A - Fibrinogen : Change prothrombin into thrombin
- (2) B - Thromboplastin : Change fibrinogen into fibrin
- (3) C - Fibrinogen : Activate thromboplastin
- (4) B - Thrombin : Change fibrinogen into fibrin

8)

How many of the following statements are correct

- (A) Systemic circulation takes place between left ventricle and right atrium of heart
- (B) Adrenal medullary hormones can decrease the cardiac output
- (C) Heart failure is not the same as cardiac arrest or heart attack
- (D) The opening between the right atrium and the right ventricle is guarded by a valve formed of two muscular flaps

- (1) Two
- (2) One
- (3) Four
- (4) Three

9)

Choose the correct statements :

- (i) A person with 'A' blood group has anti-B antibody in the plasma
- (ii) A person with 'O' blood group has no antibodies in the plasma
- (iii) A person with 'AB' blood group has no antibodies in the plasma
- (iv) A person with 'AB' blood group has both the antigens A and B on the RBC's

- (1) (i), (ii), (iii)
- (2) (i), (iii)
- (3) (i), (iii), (iv)
- (4) (ii), (iii), (iv)

10) The sino-atrial node is the normal pacemaker of the human heart because :

- (1) It is located in the right atrium
- (2) It is the only part of the conducting system that can generate impulse
- (3) It generate the maximum number of action potentials in the conducting system
- (4) It produces action potentials that are more in size than the other parts of the conducting system.

11) **Statement-1** : Erythrocytes are the most abundant all the cell in blood.

Statement-2 : A healthy adult man has on an average 7 millions to 8 millions of R.B.C/mm⁻³ of blood

- (1) Statement-1 correct and Statement-2 incorrect.
- (2) Statement-1 and Statement-2 correct.
- (3) Statement-1 incorrect and Statement-2 correct.
- (4) Statement-1 and Statement-2 incorrect.

12) **Assertion** : Heart is mesodermally derived organ situated in the thorasic cavity in between two lungs.

Reason : The atrium and ventricle of the same side are separated by thin fibrous tissue called atrio-ventricular septum.

- (1) Both assertion and reason are true and the reason is correct explanation of assertion.
- (2) Both assertion and reason are true but reason is not a correct explanation of the assertion.
- (3) Assertion is true but reason is false.
- (4) Both Assertion and reason are true.

13) **Assertion** : Persons with AB blood group are called universal recipients.

Reason : Persons with AB blood group can accept blood from persons with AB as well as the other groups of blood.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false.
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

14) **Statement-I** : Platelets are also called Leucocytes.

Statement-II : Platelets are cell fragments produced from megakaryocytes.

- (1) Both **Statement I** and **Statement II** are incorrect
(2) **Statement I** is correct but **Statement II** is incorrect
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

15)

	Column - I		Column - II
A.	Fibrinogen	i.	Osmotic balance
B.	Albumin	ii.	Defense mechanism
C.	Globulin	iii.	Blood clotting

- (1) A-iii, B-i, C-ii
(2) A-iii, B-ii, C-i
(3) A-ii, B-i, C-iii
(4) A-i, B-iii, C-ii

16) Match the columns and choose the correct option -

	Column I		Column II
A.	Resist infection	P.	Lymphocyte
B.	Heparin secretion	Q.	Neutrophil
C.	Phagocytic cell	R.	Eosinophil
D.	Immune response	S.	Basophil

	A	B	C	D
(1)	R	Q	P	S
(2)	R	Q	S	P
(3)	R	P	Q	S
(4)	R	S	Q	P

- (1) 1
(2) 2
(3) 3
(4) 4

17) Match the column-I with column-II and choose correct option :-

	Column-I		Column-II
A	Lymphocyte	(i)	2-3%
B	Basophil	(ii)	60-65%
C	Eosinophil	(iii)	0.5-1%
D	Neutrophil	(iv)	20-25%

	A	B	C	D
(1)	(iii)	(i)	(ii)	(iv)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(iv)	(i)	(ii)	(iii)

(1) 1

(2) 2

(3) 3

(4) 4

18) How many statements are correct for formed elements ?

(a) RBC + WBC + Platelets

(b) Blood - plasma

(c) Blood - serum - RBC

(d) Serum - RBC

(1) One

(2) Two

(3) Three

(4) Four

19) How many statements are correct for RBC?

(i) Biconcave and non-nucleated in human.

(ii) It contains carbonic anhydrase

(iii) It contain antigen of blood group on surface.

(iv) It contains mitochondria in large number.

(1) 2

(2) 3

(3) 1

(4) 4

20) Choose the **incorrect** one :-

(1) RBCs are devoid of nucleus in mammals and are biconcave in shape

(2) A healthy individual has 12-16 gm of haemoglobin in every 100 ml of blood

(3) RBCs are formed in the bone marrow in the all bones in the adults

- (4) Haemoglobin molecules present in RBCs play a significant role in the transport of respiratory gases

21) In the given table choose the correct value of partial pressure :-

Respiratory gas	Alveoli	Deoxygenated blood	Oxygenated blood	Tissue
O ₂	104	b	95	d
CO ₂	a	45	c	45

	a	b	c	d
(1)	40	40	40	40
(2)	45	45	95	45
(3)	45	40	45	95
(4)	40	95	40	45

- (1) 1
(2) 2
(3) 3
(4) 4

22) Neural signal from which centre can reduce the duration of inspiration ?

- (1) Chemosensitive area
(2) Respiratory rhythm centre
(3) Pneumotaxic centre
(4) Aortic body

23)

Match the column I with column II :

Column-I					Column-II	
A	Tracheal tubes				I	Scorpion
B	Gills				II	Frog
C	Moist skin				III	Insect
D	Book lungs				IV	Aquatic arthropods
	A	B	C	D		
(1)	III	IV	II	I		
(2)	III	I	IV	II		
(3)	IV	III	II	I		

(4)	IV	II	I	III
-----	----	----	---	-----

(1) 1

(2) 2

(3) 3

(4) 4

24)

CO₂ dissociates from carbaminohaemoglobin when :-

(1) P_{O₂} is low and P_{CO₂} is high

(2) P_{O₂} is low and P_{CO₂} is Low

(3) P_{O₂} is high and P_{CO₂} is low

(4) P_{O₂} is high and P_{CO₂} is high

25) The thoracic chamber is formed dorsally by the A , ventrally by the B , laterally by the C and on the lower side by the dome-shaped D .

(1) A-Sternum

(2) B-Diaphragm

(3) C-Ribs

(4) D-Vertebral column

26) Which of the following statement is/are correct?

(a) Flatworms exchange Oxygen and Carbon di oxide by simple diffusion over their entire body surface

(b) Larynx is a cartilaginous box which help in sound production

(c) Epiglottis is made up of collagen fibrous connective tissue

(d) In birds larynx is a sound producing organ

(1) only (a)

(2) only (b)

(3) (a) and (b)

(4) (a), (c) and (d)

27) How many cellular layers are present in the diffusion membrane in lungs ?

(1) one

(2) two

(3) three

(4) four

28) If total O₂ transported by blood from lungs to cell in a minute is 500 ml. Then the volume of O₂ transported by blood found dissolved in plasma is

- (1) 970 ml
- (2) 15 ml
- (3) 1000 ml
- (4) 30 ml

29) Some statements are given, read them and choose disorder :-

- (i) Allergic condition
- (ii) Difficulty in breathing
- (iii) Increased I_{gE}
- (iv) Wheezing sound

- (1) Emphysema
- (2) Anthracis
- (3) Silicosis
- (4) Asthma

30) From the following relationships between respiratory volume and capacities, mark the correct option :-

- (i) Inspiratory capacity (IC) = Tidal volume (TV) + Residual volume (RV).
- (ii) Vital capacity (VC) = Tidal volume (TV) + Inspiratory reserve volume (IRV) + Expiratory reserve volume (ERV)
- (iii) Residual volume (RV) = Vital Capacity (VC) - Inspiratory reserve volume (IRV)
- (iv) Tidal volume (TV) = Inspiratory capacity (IC) - Inspiratory reserve volume (IRV)

- (1) (i) incorrect, (ii) incorrect, (iii) incorrect, (iv) correct
- (2) (i) incorrect, (ii) correct, (iii) incorrect, (iv) correct
- (3) (i) correct, (ii) correct, (iii) incorrect, (iv) correct
- (4) (i) correct, (ii) incorrect, (iii) correct, (iv) incorrect

31) How many organism in the list given below do respiration by general body surface.
Insects, Sponges, Flatworm, Fishes, Birds, Coelenterates, Reptiles

- (1) 3
- (2) 4
- (3) 5
- (4) 6

32) Considering the following labels for mentioned structures and select the correct sequence of their arrangement :-

- A = Terminal bronchiole
- B = Respiratory bronchiole
- C = Tertiary bronchi
- D = Primary bronchi
- E = Secondary bronchi
- F = Total pulmonary bronchioles

- (1) $D \rightarrow E \rightarrow C \rightarrow F \rightarrow A \rightarrow B$

- (2) $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F$
- (3) $C \rightarrow F \rightarrow A \rightarrow B \rightarrow D \rightarrow E$
- (4) $D \rightarrow E \rightarrow F \rightarrow C \rightarrow B \rightarrow A$

33)

How much CO_2 is removed by our lungs ?

- (1) 200 ml/minute
- (2) 200 ml/hour
- (3) 18 litres/day
- (4) 200 litre/minute

34) Identify the correct sequence of events in a cardiac cycle

- (1) Joint Diastole, atrial systole, ventricular systole, ventricular diastole
- (2) Atrial systole, ventricular systole, Joint diastole, ventricular diastole
- (3) Ventricular systole, ventricular diastole, Joint diastole, atrial systole
- (4) Ventricular diastole, Joint diastole, ventricular systole, atrial systole.

35)

The contraction of which muscle is responsible for normal expiration ?

(A) Diaphragm (B) EICM

(C) IICM (D) Abdominal muscles

- (1) A, B and D
- (2) C and D
- (3) A and C
- (4) No muscles contract during normal expiration

36) Read the following statements:- (A) It is a chronic disorder

(B) Major cause of it is cigarette smoking.

(C) Alveolar walls are damaged

(D) Respiratory surface is decreased.

(E) It is due to inflammation of bronchi & bronchioles.

How many of the above statements are incorrect about "emphysema" disorder ?

- (1) One
- (2) Three
- (3) Four
- (4) All

37) P_{50} value increases when :-

- (1) CO_2 conc. increase
- (2) O_2 conc. increase

- (3) CO₂ conc. decreases
- (4) Temperature decrease

38) Which of the following is not correct ?

- (1) The nasal cavity warms and humidifies the air before it enter the lungs
- (2) The right lung is composed of three lobes, but left lung has only two lobes
- (3) lung volume and capacity measures lung functions
- (4) The visceral pleural membrane is in direct contact with the chest wall

39) Besides RBC, blood plasma also carries O₂ in solutions. The % is :-

- (1) 3.9%
- (2) 1-2%
- (3) 2-3%
- (4) 3-6%

40) A special system of blood vessels is present in our body exclusively for the circulation of blood to and from the cardiac musculature. Such type of blood circulation is known as :-

- (1) Pulmonary circulation
- (2) Systemic circulation
- (3) Coronary circulation
- (4) Double circulation

41) **Assertion :-** The pause between the end of the second sound and the beginning of the first sound coincides with ventricular diastole.

Reason :- The second sound (dub) is created by the closure of the semilunar valve.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

42) The pH of blood is :

- (1) Between 7-8
- (2) Between 2-4
- (3) Between 12-14
- (4) Between 2-5

43) In a cardiac cycle, duration of *atrial diastole* is-

- (1) 0.1 second
- (2) 0.3 seconds
- (3) 0.7 seconds

(4) 0.5 seconds

44) Centre for heart beat regulation is present is :

- (1) Pons varoli
- (2) Cerebellum
- (3) Cerebrum
- (4) Medulla

45) Read the following statement (A-D) :-

- (A) Artery always carry blood from heart to the organs
- (B) Valves are absent in the arteries
- (C) Artery always carry oxygenated blood
- (D) Lumen of artery is wide

How many statements are wrong

- (1) Three
- (2) Four
- (3) One
- (4) Two

46) Consider the following

Nucleolus, Kinetochore, Centromere, Telomere, Secondary constriction, Nuclear pore, Nucleosome, Nuclear Lamina

How many of the above are present in interphase nucleus?

- (1) Five
- (2) Four
- (3) Six
- (4) Three

47) Find incorrect statement with respect to centrosome and centrioles :-

- (a) Centrosomes are surrounded by amorphous pericentriolar material
- (b) In centrosome, both centrioles lie parallel to each other in which each has an organisation like the cartwheel
- (c) Centrioles are made up of nine unevenly spaced peripheral fibrils of tubulin
- (d) Hub is the central proteinaceous part of centriole
- (e) Proteinaceous radial spokes connect hub to peripheral triplets

- (1) a,b, e
- (2) only b
- (3) b, c
- (4) All are correct

48) (A) Unicellular organisms are capable of independent existence

- (B) Cell is the fundamental structural and functional unit of all living organisms
- (C) All cells arise from pre-existing cells
- (D) The cytoplasm is the main arena of cellular activities in all type of cells

Option :-

- (1) Statement A,C and D are wrong
- (2) Statement A, B and D are not correct
- (3) Statement A, B and C are not wrong
- (4) Statement B, C and D are not correct

49) (a) Primary cell wall is capable of growth.

(b) A non-living rigid structure called as cell wall forms an outer covering for the plasma membrane of fungi and plants.

(c) Secondary cell wall gradually diminishes as the cell matures and the primary cell wall is formed.

(d) The cell wall and middle lamellae may be traversed by plasmodesmata.

Choose the correct option for correct statements:-

- (1) a, c, and d
- (2) a, b, c and d
- (3) b, c, and d
- (4) a, b and d

50) Endomembrane system consists of organelles and these organelles perform the function.

Which of the option represent correct function of these organelles ?

- a. Formation of steroid hormones.
- b. Formation of glycoprotein.
- c. Important for osmoregulation.
- d. Storage of water.
- e. Heterophagy & Autophagy.
- f. Digestion (Intracellular & Extracellular).
- g. Aerobic Respiration.
- h. Photosynthesis.

- (1) All except a, d, g, h
- (2) All except g, h
- (3) All except d, g, h
- (4) All except a, g, h

51) **Statement-I** : The chloroplast are double membrane bound structure. Out of the two, the inner membrane is relative more permeable.

Statement-II : The ribosomes of the chloroplast are smaller then the cytoplasmic ribosomes

Choose the correct option ?

- (1) Statement-I is correct an statement-II is incorrect.
- (2) Statement-I is incorrect but statement-II is correct.
- (3) Statement-I and statement-II both are correct.
- (4) Statement-I and statement-II both are incorrect.

52) Materials to be packaged in the form of vesicles from the __ (i) __ fuse with the __ (ii) __ of the golgi apparatus and move towards the __ (iii) __.

Choose the correct terms for blanks.

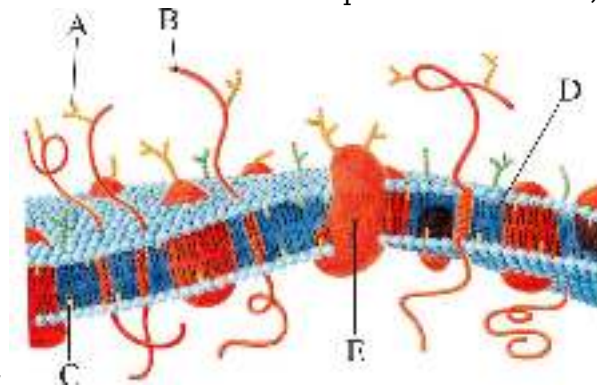
- (1) (i) - Lysosome, (ii) - Maturing face, (iii) Trans face
- (2) (i) - ER, (ii) - Cis face, (iii) Forming face
- (3) (i) - Vacuole, (ii) - Concave face, (iii) Convex face
- (4) (i) - ER, (ii) - Forming face, (iii) Trans face

53) **Assertion (A) :-** Mitochondria are the sites of aerobic respiration.

Reason (R) :- They produce ATP by the process of photophosphorylation.

- (1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.
- (2) **(A)** is correct but **(R)** is not correct.
- (3) **(A)** is incorrect but **(R)** is correct.
- (4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**.

54) The figure given below show the structure of plasma membrane, with its parts labelled from A to



E, Identify the correct :-

- (1) A → 40 percent of erythrocyte membrane
- (2) B → Movement of this measured fluidity of membrane
- (3) C → Help in facilitated transport of polar molecules
- (4) D → Its polar tail contain saturated hydrocarbon

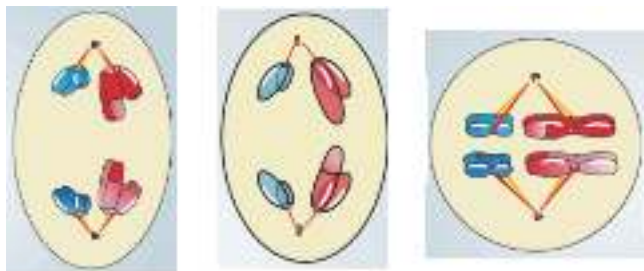
55) **Assertion :** In early prophase, chromosome which had undergone duplication during interphase begins to move towards opposite poles of the cell.

Reason : Cells at the beginning of prophase, do not show nuclear envelop, when viewed under the microscope.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

56)

Identify A, B and C in the given diagram.



	A	B	C
(1)	Meiosis-II	Meiosis-I	Metaphase
(2)	Metaphase-II	Metaphase-I	Anaphase
(3)	Anaphase-I	Anaphase	Metaphase-II
(4)	Anaphase-I	Anaphase-II	Metaphase-I

(1) 1

(2) 2

(3) 3

(4) 4

57) Which of the following statements are correct ?

(a) In animals, meiosis involves two sequential cycles of centromere splitting and nuclear division.

(b) If karyokinesis is not followed by cytokinesis, it results in multinucleated condition.

(c) During cell cycle, the cytoplasmic increase is a continuous process but DNA replication and separation of sister chromatids occurs in specific stages.

(d) Mitosis results in reduction of chromosome number by half so to conserve the specific chromosome number across generations.

(e) G_0 phase is an inactive stage because all the cells in this stage are metabolically active but never show division again.

Options :-

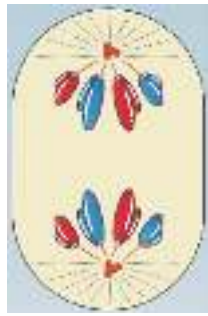
(1) a,b,c,d & e

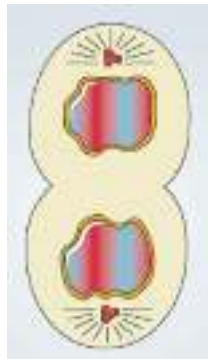
(2) b,c & e only

(3) a,b & d only

(4) b & c only

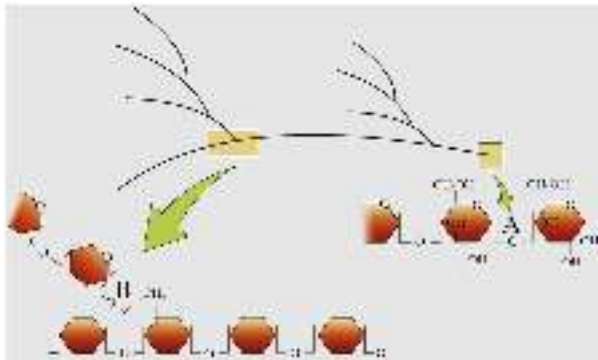
58) Identify the correct match from the column-I, II and III.

Column I		Column II		Column III	
A	Late prophase	a.		i.	ER, Golgi complex disappear

B	Anaphase	b.		ii.	Chromosome decondense and loose their individuality
C	Telophase	c.		iii.	Longest phase of meiosis-I
D	Prophase-I	d.		iv.	Splitting of centromere occur

- (1) B-a-iii, A-c-iv, C-b-i, D-d-ii
(2) A-b-i, B-a-iv, C-d-ii, D-c-iii
(3) A-d-iii, B-b-iv, C-b-i, D-d-ii
(4) B-a-iii, A-c-iv, C-a-ii, D-d-i

59) Identify A and B bonds in the following diagrammatic representation of a portion of glycogen.



- (1) A = 1-4 α -glycosidic bonds,
B = 1-6 α -glycosidic bounds.
(2) A = 1-6 α -glycosidic bonds,
B = 1-4 α -glycosidic bounds.
(3) A = 1-1 α -glycosidic bonds,
B = 1-1 α -glycosidic bounds.
(4) A = 1-4 α -glycosidic bonds,
B = 1-4 α -glycosidic bounds.

60) Read the given statements :-

- I. Amino acids are linked by peptide bond
II. Polysaccharides are linked by glycosidic bond

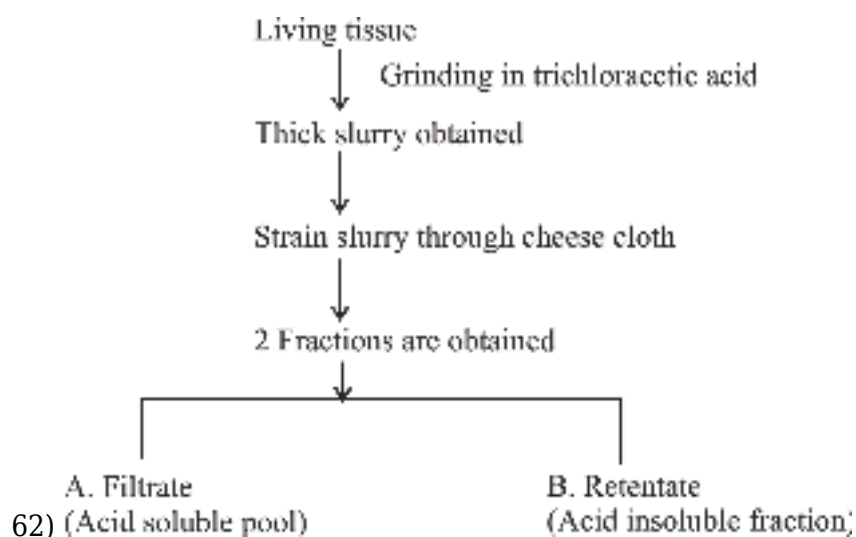
III. Only DNA work as a genetic material
 IV. Collagen is the most abundant protein in animal world
 Which of the given statement are correct

- (1) I, II, III & IV
- (2) I, & II
- (3) I, II & III
- (4) I, II & IV

61) Read the following statements and find out which is/are **correct** ?

- (A) Proteins are homopolymers of amino acids.
 (B) Amino acids are substituted ethanes.
 (C) Glycerol is also known as trihydroxy ethane.
 (D) Paper made from plant pulp is cellulose.
 (E) RuBisCO is most abundant protein in animals.

- (1) A, B, C and E
- (2) B, C and D
- (3) A and E
- (4) Only D



- I. Molecular weight ranging from less than 1000 Dalton approximately
 II. Has four type of organic compounds proteins, nucleic acids, polysaccharides and lipids
 III. Contain chemicals that have molecular weight more than 1000 Dalton
 IV. Has monomers
 V. Has generally polymers except lipid
 VI. Mostly micromolecules
 VII. Generally macromolecules

How many of the above statements (I-VII) are correct regarding A & B respectively ?

	A	B
(1)	I, II, III	IV, V, VI, VII

(2)	II, IV, VI	I, III, V, VII
(3)	I, IV, VI	II, III, V, VII
(4)	I, III, V	II, IV, VI, VII

(1) 1

(2) 2

(3) 3

(4) 4

63) Which of the following is a group of essential amino acids?

(1) Threonine, Valine, Lysine and Tryptophan

(2) Threonine, Glutamic acid, Lysine and Tryptophan

(3) Lysine, Arginine, Glycine and Methionine

(4) Lysine, Alanin, Glutamic acid and methionine

64) Find out the correct match from the following table :-

	Column I	Column II	Column III
(I)	Maltose	α - β (1-2) linkage	Hydrolysis of starch
(II)	Sucrose	α (1-4) linkage	Cane sugar
(III)	Lactose	β (1-4) linkage	Milk

(1) III only

(2) I and II only

(3) II and III only

(4) II only

65) The catalytic cycle of an enzyme action is described as:

A. Enzyme releases products of the reaction and gets free.

B. Substrate induces the enzymes to alter its shape.

C. The substrate binds with the active site of the enzymes.

D. Enzymes-product complex is formed.

Choose the **correct** answer from the options given below:

(1) C $\rightarrow$ B $\rightarrow$ D $\rightarrow$ A

(2) B $\rightarrow$ C $\rightarrow$ D $\rightarrow$ A

(3) D $\rightarrow$ C $\rightarrow$ A $\rightarrow$ B

(4) C $\rightarrow$ B $\rightarrow$ A $\rightarrow$ D

66) Select correct & incorrect statements-

(i) There could be many more altered structural states between the stable substrate and product.

(ii) Stability is something related to energy status of the molecule or structure.

(iii) If 'P' is at lower level than 'S' the reactions on endothermic reaction.

(iv) Activity of enzyme can be affected by change in the conditions which can alter the tertiary structure of enzyme.

- (1) Correct statements-3, incorrect-1
 (2) Correct statements-2, incorrect-2
 (3) Correct statements-1, incorrect-3
 (4) Correct statements-4

67)

Choose correct option to match column-A and column-B

Column - A		Column - B	
(i)	Low temperature	(a)	Temporarily inactive state of enzyme
(ii)	Destruction of enzymatic activity	(b)	Maximum activity of enzyme
(iii)	$V_{\max}$	(c)	High temperature
(iv)	Competitive inhibitor	(d)	Closely resemble to substrate

- (1) i-a, ii-c, iii-b, iv-d
 (2) i-a, ii-c, iii-d, iv-b
 (3) i-c, ii-b, iii-d, iv-a
 (4) i-a, ii-b, iii-c, iv-d

68) Which of the following are true for polysaccharides?

- (i) It is acid insoluble.
 (ii) Polysaccharides are long chains of sugars (monosaccharides).
 (iii) Cellulose is a polymeric polysaccharide consisting of only one type of monosaccharide.
 (iv) Starch is a polymeric polysaccharide present as a store house of energy in plant tissues.
 (v) Glycogen is a heteropolymer and not a homopolymer.
 (vi) Animals and plants have polymeric polysaccharide called glycogen.

- (1) (i), (ii), (iii), (iv) and (v)
 (2) (i), (ii), (iii) and (iv)
 (3) (i), (ii), (iii), (iv) and (vi)
 (4) All are correct

69) Cells having 20 bivalents undergoes meiosis I then calculate the number of chromatids in each nuclei after meiosis I.

- (1) 20
 (2) 40
 (3) 10
 (4) 80

70) Meiosis ensures the production of _____ phase in the life cycle of sexually reproducing organisms, whereas fertilisation restores the _____ phase.

- (1) Haploid; diploid
- (2) haploid : haploid
- (3) diploid; haploid
- (4) diploid; diploid

71) If a meiocyte in G_1 stage contains 40 picogram (pg) of DNA, then what will be the amount of DNA in the products of meiosis II?

- (1) 20 pg
- (2) 40 pg
- (3) 10 pg
- (4) 80 pg

72) Identify the stage with correct description.

- (1) Diakinesis-The compaction of chromosomes begin in this stage
- (2) Diplotene - the beginning of this stage is recognised by the dissolution of the synaptonemal complex.
- (3) Leptotene- Chromosomes start pairing together and his process is called synapsis
- (4) Pachytene-It represents transition to metaphase-I

73) Phragmoplast is formed by Golgi complex and grows

- (1) centripetally to from cell plate.
- (2) centrifugally to from cell plate.
- (3) centripetally to produce a cell furrow.
- (4) centrifugally to form a cell furrow.

74) In the female gamete of a plant cell has four chromosomes, then its parental cell will have

- (1) 16 chromosomes during anaphase-I
- (2) 4 bivalents aligned on the equatorial plate during metaphase-I
- (3) 16 tetrads during pachytene.
- (4) 16 chromosomes during G_2 stage.

75) Which of the following statement is true about meiosis?

- (a) It involves two sequential cycle of nuclear and cells division called meiosis I and II but only a single cycle of DNA replication.
- (b) It involves pairing of homologous chromosome and recombination between them.
- (c) Four haploid cells are formed at the end of meiosis.

- (1) Only (a) & (b) are correct
- (2) Only (c) is correct
- (3) Only (a) & (c) are correct
- (4) All are correct

76) Which of the following is not a feature of telophase?

- (1) Chromosomes cluster at opposite poles
- (2) Centromere split and chromatids separate.
- (3) Nuclear envelope assembles around the chromosome clusters.
- (4) Nucleolus, Golgi complex and ER reform.

77) Identify the wrong statement regarding metaphase of mitosis.

- (1) Condensation of chromosomes is completed
- (2) Chromosome is made up of chromatids, which are held together by the chiasmata
- (3) Small disc-shaped structures present at the surface of the centromeres are called kinetochores.
- (4) Chromosomes are moved to spindle equator and get aligned along metaphase plate through spindle fibres.

78) Select the correct match :

- (1) Ribosomes - Non-membranous and made up of DNA and proteins
- (2) Golgi apparatus - Important site for formation of glycolipids and glycoproteins
- (3) Secondary cell wall - Capable of extension growth
- (4) Endomembrane system - Functions not co-ordinated

79) (a) Granular structure.

(b) First observed under the electron microscope by George Palade.

(c) Composed of RNA and proteins.

(d) Not surrounded by any membrane.

Above given statements are true for which cell organelle ?

- (1) Nucleolus
- (2) Ribosomes
- (3) Cristae
- (4) Chloroplast

80) Select the correct statement from the following

- (1) Inorganic catalysts work efficiently at high temperature and high pressure while enzymes get damaged at high temperature (above 40°C).
- (2) Ribozymes are Ribo-nucleic acids behaving like enzymes.
- (3) Thermophilic organisms living in hot vents and sulphur springs have enzymes that are stable and retain their catalytic power even at high temperature (80-90°C)
- (4) All are correct

81) An enzyme acts by

- (1) reducing the energy of activation
- (2) increasing the energy of activation

- (3) decreasing the pH
- (4) increasing the pH

82) A competitive inhibitor of succinic dehydrogenase is

- (1) α -ketoglutarate
- (2) malate
- (3) malonate
- (4) oxaloactate

83) Which of the following is a co-factor for the proteolytic enzyme carboxypeptidase?

- (1) Zinc
- (2) Copper
- (3) Calcium
- (4) Magnesium

84) Glycosidic linkage at place of branching in starch and glycogen is

- (1) $\alpha(1 \rightarrow 6)$
- (2) $\alpha(1 \rightarrow 4)$
- (3) $\beta(1 \rightarrow 4)$
- (4) $\beta(1 \rightarrow 6)$

85) Which one of the following is the correct sequence of carbohydrates in the order of increasing complexity of chemical structure.

- (1) Sucrose, Starch, Oligosaccharide, Maltose, Triose
- (2) Triose, Maltose, Sucrose, Oligosaccharide, Starch
- (3) Triose, Glucose, Maltose, Oligosaccharide, Starch
- (4) Oligosaccharide, Triose, Starch, Glucose, Maltose

86) Select the unsaturated fatty acid having 20 carbons.

- (1) Palmitic acid
- (2) Palmitoleic acid
- (3) Arachidonic acid
- (4) Linolic acid

87) In animal world, the most abundant protein is _____. while in the whole biosphere the most abundant protein is _____.

- (1) Antibody; Collagen
- (2) Collagen; RuBisCO
- (3) RuBisCO; Collagen
- (4) Collagen; Oxidase

88)

A protein is a sequence of amino acid, in which first and last amino acids are called respectively.

- (1) N-terminal and C-terminal amino acids.
- (2) C-terminal and N-terminal amino acid.
- (3) α and β -amino acid.
- (4) β and α -amino acid.

89) Select the correct statement which respect to prophase-I

A	Longest chromosomes are seen in leptotene
B	Zygotene is also called bouquet stage
C	Synaptonemal complex is formed between homologous chromosomes
D	Recombinase enzyme is necessary for recombination process
E	Nucleolus disappears in diakinesis

Choose the correct answer from the options given below :

- (1) A, B, C and E only
- (2) A, C, D and E only
- (3) A and D only
- (4) B, C, D and E only

90) A plant has 10 pg DNA and 8 chromosomes in its pollen grain, then what was the amount of DNA and number of chromosomes in its microspore mother cell at G_1 phase?

- (1) 40 pg ; 16 chromosomes
- (2) 20 pg ; 16 chromosomes
- (3) 10 pg ; 8 chromosomes
- (4) 20 pg ; 32 chromosomes

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	2	2	2	3	4	3	2	1	3	4	2	3	4	4	2	2	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	1	4	2	4	1	2	2	2	3	1	3	1	3	2	2	1	2	3
Q.	41	42	43	44	45															
A.	4	2	1	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	3	2	4	4	1	3	1	1	2	3	4	1	3	4	3	2	3	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	3	2	3	2	2	3	1	3	2	1	2	3	4	4	4	4	4	3
Q.	86	87	88	89	90															
A.	4	2	2	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	2	3	3	4	4	4	1	3	3	1	3	4	3	1	4	3	2	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	1	3	3	3	2	2	4	2	1	1	1	1	4	1	1	4	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	3	4	4	2	3	3	4	2	2	4	2	2	4	4	4	2	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	1	1	1	1	1	2	2	1	1	2	2	2	4	2	2	2	2	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	1	1	3	3	2	1	2	2										

SOLUTIONS

PHYSICS

$$1) n_p = \frac{v}{4L}$$

$$n_q = \frac{v}{2L}$$

$$n_r = \frac{2v}{3L}$$

$$n_s = \frac{3v}{4L}$$

Ratio :

$$n_p : n_q : n_r : n_s = 1 : 2 : 3 : 4$$

$$2) f_t - 5 = \frac{v}{2 \times 21} \quad \dots(1)$$

$$f_t + 5 = \frac{v}{2 \times 20} \quad \dots(2)$$

$$\Rightarrow 2(f_t + 5)20 = 2(f_t - 5)21$$

$$\Rightarrow 20f_t + 100 = 21f_t - 105$$

$$\Rightarrow f_t = 205 \text{ Hz}$$

$$3) n_A = 256$$

As tuning fork A when sounded with tuning fork B gives four beats, therefore the frequency n_B of tuning fork B is,

$$n_B = 256 \pm 4 = 252 \text{ or } 260$$

When the tuning fork A is slightly loaded with wax, the frequency of A decreases and the difference between two frequencies decreases if

$$n_B = 252$$

As on sounding beats decreases, hence

$$n_B = 252 \text{ Hz.}$$

4) After 2 s, both the pulses reach 4 cm distance towards each other and thus they appear one over the other. Since their amplitudes are equal and opposite so the resultant pulse is of zero amplitude, which means that the string will be straight and thus there will be no potential energy in the string and its total energy will be purely kinetic.

5)

Explain question : Find number of beats per second and intensity ratio between waxing and waning due to two super posed sound waves.

Concept: This question is based on beat frequency and intensity of wave.

Solution:

Formula: $f_{\text{beat}} = |f_1 - f_2|$ and $I = \left(\frac{A_{\text{max}}}{A_{\text{min}}}\right)^2$

Calculation:

$$\text{Number of beats} = \frac{508\pi}{2\pi} - \frac{500\pi}{2\pi} = 4$$

$$\text{and } \frac{I_{\text{max}}}{I_{\text{min}}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{3+6}{3-6}\right)^2 = \frac{9}{1}$$

Final answer: (2)

6) beat frequency $\Delta n = 4\text{Hz}$

Bent time period = $\frac{1}{4}$ sec

For maximum to minimum $t = \frac{T_b}{2}$

$t = \frac{1}{8}$ sec

7) $y = 5 \sin\left(\frac{2\pi x}{3}\right) \cos 20\pi t$, comparing with equation

$y = 2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi vt}{\lambda} \Rightarrow \lambda = 3$, distance

between two adjacent nodes = $\lambda/2 = 1.5$ cm.

8) $f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$

If radius is doubled and length is doubled, mass per unit length will become four times. Hence

$f' = \frac{1}{2 \times 2\ell} \sqrt{\frac{2T}{4\mu}} = \frac{f}{2\sqrt{2}}$

9) $y = 20 \sin 2\pi (100t) \cos 2\pi (0.02x)$

$\square A = 20$, loop length = $\lambda/2 = \frac{50}{2} = 25$ units

10)

Third overtone of closed organ pipe means seventh harmonic.

$(n_7)_{\text{closed}} = (n_4)_{\text{open}}$

$$\Rightarrow 7 \left(\frac{v}{4\ell_c}\right) = 4 \left(\frac{v}{2\ell_0}\right) = \frac{\ell_c}{\ell_0} = \frac{7}{8}$$

11) $v = n\lambda$

$300 = 500 \lambda \Rightarrow \lambda = 60$ cm

$$\begin{array}{l}
 \text{First resonance} = \frac{\lambda}{4} = 15 \text{ cm} \\
 \text{Second resonance} = \frac{3\lambda}{4} = 45 \text{ cm} \\
 \text{Third resonance} = \frac{5\lambda}{4} = 75 \text{ cm}
 \end{array} \left. \vphantom{\begin{array}{l} \text{First resonance} \\ \text{Second resonance} \\ \text{Third resonance} \end{array}} \right\} \rightarrow 2 \text{ Resonance possible}$$

Minimum liquid level = $80 - 75 = 5 \text{ cm}$
 Maximum liquid level = $80 - 15 = 65 \text{ cm}$

12) For closed pipe frequencies will be,
 $= f_0, 3f_0, 5f_0, 7f_0$
 $= 85, 255, 425, 595$

13)

A. Concept: The ideal gas law, $PV = nRT$, governs the behavior of ideal gases. Isotherms represent the relationship between pressure (P) and volume (V) at constant temperature (T). For a fixed mass of gas (constant n), the equation can be rearranged to understand the

relationship between $\frac{1}{V}$, P, and T. Crucially, the slope of the $\frac{1}{V}$ vs. P graph is related to the temperature.

B. Mathematical Calculation:

1. Ideal Gas Law: $PV = nRT$

2. Rearranging for $\frac{1}{V}$: $\frac{1}{V} = \left(\frac{nR}{T} \right) P$

3. Slope: In this equation, if we plot $\frac{1}{V}$ on the y-axis and P on the x-axis, the term $\left(\frac{nR}{T} \right)$ represents the slope of the line.

4. Slope and Temperature: Since n and R are constant, the slope is inversely proportional to the

temperature T: Slope $\propto \frac{1}{T}$

5. Analyzing the Graph: The graph shows three lines (isotherms) for temperatures T_A , T_B , and T_C . We observe the slopes of these lines.

Observation from Graph: From the graph, we see:

Slope A > Slope B > Slope C

Relating Slope to Temperature: Since the slope is inversely proportional to temperature, this means $T_A < T_B < T_C$

C. Final Answer: $T_A < T_B < T_C$

Option 2

14) $\mu = \mu_1 + \mu_2$

$$\begin{aligned}
 \frac{P(2V)}{RT_1} &= \frac{P'V}{RT_1} + \frac{P'V}{RT_2} \Rightarrow \frac{2P}{RT_1} = \frac{P'}{R} \left[\frac{T_2 + T_1}{T_1 T_2} \right] \\
 P' &= \frac{2PT_2}{(T_1 + T_2)} = \frac{2 \times 1 \times 600}{(300 + 600)} = \frac{4}{3} \text{ atm}
 \end{aligned}$$

15)

$$PT = \text{constant}$$

$$\text{compare with } \frac{T^n}{P^{n-1}} = \text{constant}$$

$$\Rightarrow PT^{\frac{n}{1-n}} = \text{constant}$$

$$\frac{n}{1-n} = 1, \text{ then } n = \frac{1}{2}$$

$$C_{\text{poly}} = C_V + \frac{R}{1-n}$$

$$\Rightarrow 4.5R = \frac{f}{2}R + \frac{R}{1-\frac{1}{2}} \Rightarrow f = 5$$

16) From Wien's displacement law, temperature (T) $\propto 1/\lambda_m$ (where λ_m is the maximum wavelength). Thus temperature of a body is inversely proportional to the wavelength. Since blue star has smaller wavelength and red star has maximum wavelength, therefore blue star is at higher temperature than red star.

$$17) T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_1 (A \ell_1)^{\gamma-1} = T_2 (A \ell_2)^{\gamma-1}$$

$$\frac{T_1}{T_2} = \left(\frac{\ell_2}{\ell_1}\right)^{\gamma-1} = \left(\frac{\ell_2}{\ell_1}\right)^{\frac{5}{3}-1} = \left(\frac{\ell_2}{\ell_1}\right)^{2/3}$$

$$18) \frac{dP}{dV} = -\frac{P}{V}$$

$$\frac{\Delta P}{\Delta V} = -\frac{P}{V}$$

$$\frac{P}{P} = -\frac{\Delta V}{V}$$

19) By work energy theorem :

$$F \times 0.2 - 0.1 \times 10 \times 2.2 = 0$$

$$F = 22 \text{ N}$$

20)

As acceleration down the inclined plane is zero,

$$f = mg \sin \theta$$

While moving up the plane

Work done by friction = $-(mg \sin \theta)l$

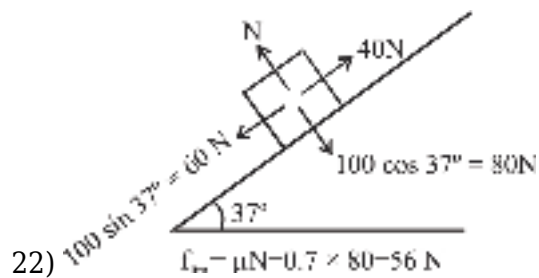
and work done by gravity = $-(mg \sin \theta)l$

$$\square W_{\text{net}} = -2mg l \sin \theta$$

$$\Delta KE = 0 - \frac{1}{2}mu^2 = -\frac{1}{2}mu^2$$

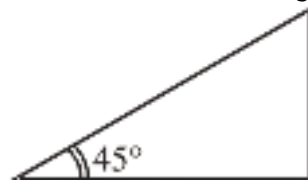
$$\square \frac{1}{2}mu^2 = 2mgl \sin \theta \Rightarrow l = \frac{u^2}{4g \sin \theta}$$

21) Impulse = Change in momentum
 $= m(v_2 - v_1) = 2[0 - 5]$
 $= -10 \text{ kg m/s}$



Net driving force = $60 - 40 = 20 \text{ N}$ (down the plane) As resisting force is greater than net driving force, the friction will be static of nature and friction force is 20 N (up the plane)

23) Time in describing a distance S on rough inclined plane is given by $S = \frac{1}{2} a_1 t_1^2$



and the time for smooth plane $S = \frac{1}{2} a_2 t_2^2$

$$\square \frac{1}{2} a_1 t_1^2 = \frac{1}{2} a_2 t_2^2 \text{ or } \frac{a_1}{a_2} = \frac{t_2^2}{t_1^2}$$

$$\frac{a_1}{a_2} = \frac{1}{n^2}$$

Given $nt_2 = t_1$, hence

Now acceleration down the rough plane

$$a_1 = g(\sin \theta - \mu \cos \theta) = \frac{g}{\sqrt{2}}(1 - \mu)$$

and acceleration on the smooth plane

$$a_2 = g \sin \theta = \frac{g}{\sqrt{2}}$$

$$\square \frac{a_1}{a_2} = (1 - \mu) \text{ or } (1 - \mu) = \frac{1}{n^2} \text{ or } \mu = \left(1 - \frac{1}{n^2}\right)$$

24)

Area under (F-t) curve = $M(v_f - v_i)$

$$\text{or } 50 \times 5 + 75 \times 5 = 16(v_f - 0)$$

$$\square v_f = 39 \text{ m/s.}$$

25) When particle moves away from the origin then at position $x = x_1$ force is zero and at $x > x_1$ force is positive (repulsive in nature) so particle moves further and does not return back to original position.

i.e., the equilibrium is not stable.

Similarly at position $x = x_2$ force is zero and $x > x_2$, force is negative (attractive in nature)

So particle return back to original position i.e., the equilibrium is stable.

$$\begin{aligned}
 W &= \int_0^5 F dx = \int_0^5 (7 - 2x + 3x^2) dx \\
 26) \quad &= [7x - x^2 + x^3]_0^5 \\
 &= 35 - 25 + 125 = 135 \text{ J}
 \end{aligned}$$

$$27) F = ma = \frac{mv}{T} \left(a = \frac{v-0}{t} \right)$$

Instantaneous power

$$Fv = mav = \frac{mv}{T} at = \frac{mv}{T} \frac{v}{T} t = \frac{Mv^2}{T^2} t$$

28) Here only gravity is doing its work in direction of motion.

$$(WD)_g = mgh = 2 \times 10 \times 10 = 200 \text{ J.}$$

W.D. by air drag is -ve as air drag acts in opp. direction of motion.

$$(W.D.) \Rightarrow F.D.$$

$$\Rightarrow -10 \times 10 = -100 \text{ J.}$$

net (W.D) :-

$$= 2(WD)_g - (WD)_{\text{air drag}}$$

$$= 200 - 100$$

$$= 100 \text{ J.}$$

now applying work energy theorem

$$WD = \Delta KE$$

$$100 \text{ J} = \frac{1}{2} m (V_f^2 - V_i^2)$$

$$V_f \Rightarrow \sqrt{100}$$

$$\Rightarrow 10 \text{ m/s}$$

$$29) a = \frac{P}{mv}$$

$$\frac{v dv}{dx} = \frac{P}{mv}$$

$$\int_{v_1}^{v_2} v^2 dv = \frac{P}{m} \int_0^x dx$$

$$\frac{1}{3} (v_2^3 - v_1^3) = \frac{P}{m} (x - 0)$$

$$x = \frac{m}{3P} (v_2^3 - v_1^3)$$

$$30) Mg(h + x) - F_{AV}x = 0 - 0$$

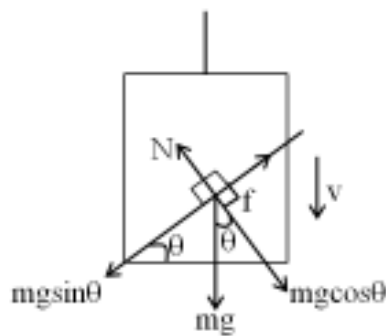
$$F_{AV} = Mg \left(\frac{h}{x} + 1 \right)$$

$$31) \text{ As } F = -\frac{dy}{dx} \text{ i.e., slope of U-X graph}$$

So, equilibrium positions are where slope will be zero.

i.e., point A,C,D,E potential energy will be minimum.

$$\begin{aligned}
 32) W_{\text{friction}} + W_{\text{spring}} &= KE_f - KE_i \\
 (\mu mg)(x)(-1) + \frac{1}{2}K(0^2 - x^2) &= 0 - \frac{1}{2}mu^2 \\
 -0.1 \times 100x - \frac{1}{2} \times (100)x^2 &= -\frac{1}{2} \times 10(25) \\
 50x^2 + 10x - 125 &= 0 \\
 10x^2 + 2x - 25 &= 0 \\
 \text{on solving } x &\approx 1.5 \text{ m}
 \end{aligned}$$



$$\begin{aligned}
 33) \text{ Block is at rest so} \\
 f &= mg \sin \theta \\
 W_f &= |f| |s| \cos(90^\circ + \theta) \\
 W_f &= (mg \sin \theta)(vt)(-\sin \theta) \\
 W_f &= -mg \sin^2 \theta (vt) \quad [\text{ } s = vt] \\
 W_f &= -mg vt \sin^2 \theta
 \end{aligned}$$

$$\begin{aligned}
 34) \frac{dw}{dt} &= P = 2x \\
 FV &= 2x \\
 \left(mv \frac{dv}{dx}\right)(v) &= 2x \\
 \int_0^v v^2 dv &= \left(\frac{2}{m}\right) \int_0^x x dx \\
 \frac{v^3}{3} &= \frac{2}{m} \frac{x^2}{2} \\
 v &= \left(\frac{3}{m}x^2\right)^{1/3}
 \end{aligned}$$

$$35) y_{\text{cm}} = \frac{m(0) + m(0) + m(b) + m(b) + m(2b)}{5m}$$

$$= \frac{4b}{5}$$

36)

$$\text{Range (R)} = \frac{u^2 \sin 2\theta}{g} = \frac{40 \times 40 \sin(2 \times 45^\circ)}{10} = 160\text{m}$$

Path of COM will not change so $X_{\text{COM}} = 160\text{ m}$

$$X_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$160 = \frac{4(200) + 2(u)}{u + 2}$$

$$960 = 800 + 2x$$

$$x = 80\text{m}$$

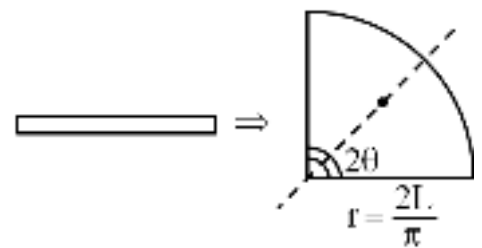
$$37) L = \frac{2\pi r}{4}$$

$$r = \frac{2L}{\pi}$$

$$\text{Com} = \frac{r \sin \theta}{\theta}$$

$$\Rightarrow \frac{2L}{\pi} \frac{\sin 45^\circ}{45^\circ \times \frac{\pi}{180^\circ}}$$

$$\Rightarrow \frac{\frac{2L}{\pi} \times \frac{1}{\sqrt{2}}}{\left(\frac{\pi}{4}\right)} \Rightarrow \left(\frac{4\sqrt{2}L}{\pi^2}\right)$$



38) **Explanation:**

Two masses, m and $3m$, are connected by a string in an Atwood machine. When released, we need to find the acceleration of the system's center of mass (COM).

Concept:

The center of mass accelerates as the total external force (due to the difference in gravitational forces) acts on the combined system of masses.

Formula:

Acceleration of each block is

$$a = \frac{(3m - m)}{3m + m} g = \frac{g}{2}$$

$$\vec{a}_{\text{cm}} = \frac{3m\vec{a}_1 + m\vec{a}_2}{3m + m}$$

Both mass have same magnitude of acceleration but in opposite direction $\vec{a}_1 = -\vec{a}_2 = a$ Let)

$$a_{\text{cm}} = \left(\frac{3m - m}{4m}\right) \times \frac{g}{2} = \frac{g}{4}$$

39)

Center of mass does not change these position ($F_{\text{ext}} = 0$)

$$\Delta x = \frac{90 + 12 - 40 \times 12}{90 + 40 \times 20} = 4$$

$$40) x = 3t - 4t^2 + t^3$$

$$V = 3 - 8t + 3t^2$$

$$\text{at } t = 0 \quad V_i = 3$$

$$\text{at } t = 4s, \quad V_f = 3 - 8(4) + 3(4)^2 = 19$$

$$W = \frac{1}{2}m(V_f^2 - V_i^2) = \frac{1}{2} \times 3 \times 10^{-3} [(19)^2 - (3)^2]$$

$$= 1.5 \times 10^{-3} \times 22 \times 16 = 528 \text{ mJ}$$

$$41) n_2 = 1 \left[\frac{1\text{kg}}{10\text{kg}} \right] \left[\frac{1\text{m}}{10\text{m}} \right]^2 \left[\frac{1\text{sec}}{1\text{min}} \right]^{-2}$$

$$= 36 \times 10^{-1}$$

$$42) y = \frac{a}{t^2} e^{-\left[\frac{t-x}{b-c}\right]}; \left[\frac{t}{b}\right] = \left[\frac{x}{c}\right] = M^0 L^0 T^0$$

$$\Rightarrow [b] = T \text{ \& } [c] = [L] \text{ \& } \left[\frac{a}{t^2}\right] = y \Rightarrow [a] = [LT^2] \text{ \& } e^{-\left(\frac{t-x}{b-c}\right)}$$

is dimensionless

43)

Different dimensions can't be added or subtracted.

$$44) R = \frac{V}{I} \Rightarrow \frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\Rightarrow \frac{\Delta R}{R} \times 100 = \left(\frac{5}{100} \times 100 + \frac{0.2}{10} \times 100 \right) = 7\%$$

$$45) LC = 1\text{MSD} - 1\text{VSD}$$

$$LC = 1\text{MSD} - \frac{8}{10}\text{MSD}$$

$$LC = 1 - \frac{8(1)}{10} = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$$

CHEMISTRY

46)

Question Explanation:

To determine order of concentration of solution in given graph.

Concept: $V.P. \propto \frac{1}{C_i}$; $T_b \propto C_i$

In given graph solution-I is less concentrated and solution-III is more concentrated because V.P. of solution-I is less decreases as compared to solution-III. So order to concentrated solution is $I < II < III$.

Final Answer : (1) $I < II < III$

47)

$$\begin{aligned}P_T &= P_A^{\circ} \times A + P_B^{\circ} \times B \\&= 120 \times 0.5 + 80 \times 0.5 \\&= 100 \text{ Torr} \\y_B &= \frac{80 \times 0.5}{100} = 0.4\end{aligned}$$

48)

Explanation :

Calculation for weight of the solute.

Given Data

- A. Molality (m) = 0.5 mol/kg
- B. Mass of solution = 2.06 kg
- C. Molar mass of urea (NH_2CONH_2) = 60 g/mol

Concepts

- A. Molality (m) = moles of solute / mass of solvent (kg)
- B. Mass of Solution = Mass of Solute + Mass of Solvent

Calculations

A. Mass of solvent:

- A. Let mass of solvent be 'x' kg.
- B. Moles of urea = 0.5x
- C. Mass of Urea = 0.5x $\times$ 60 = 30x grams = 0.03x Kg
- D. Mass of solution = x + 0.03x = 2.06 Kg
- E. 1.03x = 2.06
- F. x = 2.06/1.03 = 2 Kg
- G. Mass of solvent = 2 Kg

B. Mass of urea:

- A. Mass of solution = Mass of solvent + Mass of urea.
- B. Mass of urea = 2.06 Kg - 2 Kg = 0.06 Kg
- C. Mass of urea = 0.06 Kg $\times$ 1000 g/Kg = 60 g

Answer option 3, (60 g).

49) Concept:

Boiling Point Elevation : $\Delta T_b = i \times K_b \times m$

Solution: 1. Calculate the elevation in boiling point (ΔT_b): $\Delta T_b = i \times K_b \times m = 2 \times 0.52 \text{ K kg mol}^{-1} \times 0.1 \text{ mol kg}^{-1} = 1.04 \text{ K}$ **2. Calculate the boiling point of the solution:** Boiling point of solution = Boiling point of pure water + $\Delta T_b = 100^\circ\text{C} + 1.04^\circ\text{C} = 101.04^\circ\text{C}$

50) A. Question Explanation:

The question provides the freezing point depression of an aqueous solution and asks for the corresponding elevation in boiling point. We are given the cryoscopic constant (K_f) and the

ebullioscopic constant (K_b) for water.

B. Given Data:

- A. Freezing point depression (ΔT_f) = 0.186°C
- B. $K_f = 1.86^\circ\text{C kg/mol}$
- C. $K_b = 0.512^\circ\text{C kg/mol}$

C. Concept:

The freezing point depression and boiling point

$$\Delta T_f = K_f \times m$$

A. $\Delta T_b = K_b \times m$

Solution:

- 1. Calculate the molality (m):** $m = \Delta T_f / K_f$ $m = \frac{0.186^\circ\text{C}}{1.86^\circ\text{C kg/mol}}$ $m = 0.1 \text{ mol/kg}$
- 2. Calculate the boiling point elevation (ΔT_b):** $\Delta T_b = K_b \times m$ $\Delta T_b = 0.512^\circ\text{C kg/mol} \times 0.1 \text{ mol/kg}$ $\Delta T_b = 0.0512^\circ\text{C}$

E. Final Answer:

The elevation in boiling point is 0.0512°C . So the answer is 4.

51)

For ideal solution

$$\Delta G_{\text{mix}} < 0 \quad \Delta S_{\text{mix}} > 0$$

$$\Delta H_{\text{mix}} = 0 \quad \Delta V_{\text{mix}} = 0$$

52)

Case I: For 2 M CH_3COOH in benzene, complete dimerization means 2 molecules form 1

$$\text{Effective concentration} = \frac{2}{2} = 1\text{M}$$

dimer:

Case II: For 0.5 M KCl in water, it ionizes completely

into 2 ions (K^+ and Cl^-), so $i = 2$: $\text{Effective concentration} = 2 \times 0.5 = 1\text{M}$ Since both have an effective solute concentration of 1 M, both solutions exert the same osmotic pressure and are isotonic.

53)

$$\pi = i CRT$$

$$\pi \propto i \text{ (if } C \text{ is same)}$$

for KNO_3 $i = 2$

for CH_3COOH $1 < i < 2$

54) **Assertion** is correct, this is directly related to Henry's law, which states that the solubility of gas in a liquid is directly proportional to the partial pressure of that gas above the liquid at a constant temperature.

Reason is incorrect, this is directly related to Henry's law. The solubility of a gas in a liquid (which is inversely related to Henry's constant) increases with increasing pressure, meaning the Henry's constant itself remains constant at a given temperature for a specific gas-liquid

system.

Correct Ans. 1

55) **Question Explanation:** Which of the following mixtures will show a positive deviation.

Concept : This question is based on Raoult's law

Solution :

1. Methanol has strong H-bonds. Acetone can form weaker H-bonds with methanol than methanol by itself, leading to positive deviation.
2. Chloroform and acetone form stronger H-bonds than pure acetone, showing negative deviation.
3. Nitric acid and water form strong H-bonds, showing negative deviation.
4. Phenol and aniline form strong H-bonds, showing negative deviation.

Final Answer : option (1)

56)

Given Data:

- Both solutions have the same molality (0.1 m).
- Glucose and urea are non-electrolytes.

Concept : Elevation in Boiling point (ΔT_b)

Mathematical Calculation:

A. Elevation of Boiling Point (ΔT_b):

$$\Delta T_b = K_b \cdot m$$

A. where ΔT_b is the elevation of boiling point, K_b is the molal elevation constant, and m is the molality of the solute.

B. Since both glucose and urea are non-electrolytes and have the same molality (0.1 m) in their respective solutions:

A. The number of solute particles in both solutions is the same.

B. Therefore, according to colligative properties, they should exhibit the same elevation of boiling point.

E. Final Answer:

A. Statement-I is correct.

B. Statement-II is incorrect.

$$57) \pi_{K_2SO_4} = \pi_{NaCl}$$

$$i_1 \times C_1 = i_2 \times C_2$$

$$i_1 \times \frac{17.4}{174 \times 100} \times 1000 = 2 \times \frac{5.85}{58.5 \times 100} \times 1000$$

$$i_1 = 2$$



$$1-\alpha \qquad 2\alpha \qquad \alpha$$

$$1 + 2\alpha = 2$$

$$2\alpha = 1$$

$$\alpha = 0.5 \text{ or } 50\%$$

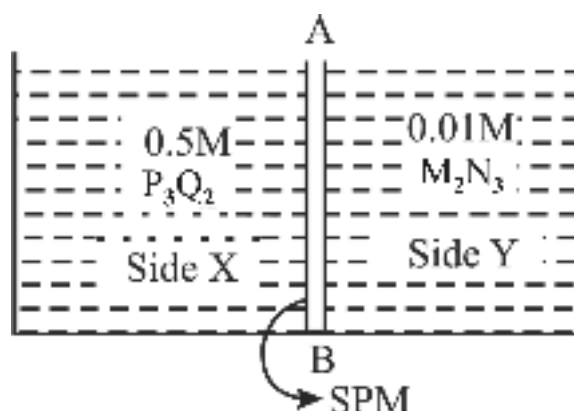
58) **Explain Question** - In this question, to determine where the brown color will appear based on the movement of water or ions.

Concept - This question based on Osmosis.

Solution -

Since the semipermeable membrane allows only water, P_3Q_2 and M_2N_3 cannot mix directly.

This means the reaction cannot occur, and the brown color cannot form.



Final Answer - Option (4)

59)

Explain Question - Which of the following is correct regarding depression in freezing point.

Concept - This question based on colligative property.

Solution -

(a) Vapour pressure of solution is less than that of pure solvent: This is correct. The presence of solute particles reduces the number of solvent molecules at the surface, thus lowering the vapor pressure of the solution compared to the pure solvent.

(d) Only solvent molecules solidify at the freezing point: This is also correct. When a solution freezes, typically only the solvent crystallizes out, leaving the solute in the remaining liquid phase.

Final Answer - Option (1)

60)

Explaining Question:

Find the van't Hoff factor (i) from given freezing point depression data.

Concept: This question is based on Colligative properties - Freezing point depression

Solution:

$$\Delta T_f = i \times K_f \times (\text{molality})$$

Given:

$$\Delta T_f = 2 \text{ K}$$

$$K_f = 1.72 \text{ K} \cdot \text{kg} \cdot \text{mol}^{-1},$$

Mass of solute = 20 g,

$$\text{Molar mass} = 172 \text{ g/mol} \rightarrow \text{moles} = 20/172 \approx 0.116 \text{ mol}$$

$$\text{Mass of solvent (benzene)} = 50 \text{ g} = 0.05 \text{ kg}$$

$$\text{Molality} = 0.116 / 0.05 = 2.32 \text{ mol/kg}$$

Now,

$$2 = i \times 1.72 \times 2.32$$

$$i = 2 / (1.72 \times 2.32) \approx 2 / 3.99 \approx 0.5$$

61) In positive deviation v.p. $\uparrow$ B.P $\downarrow$ therefore



62) Bond strength $\propto$ Bond order

removal of electron from antibonding MO increases B.O.

NO & O₂ has valence e⁻ in π^* orbital.

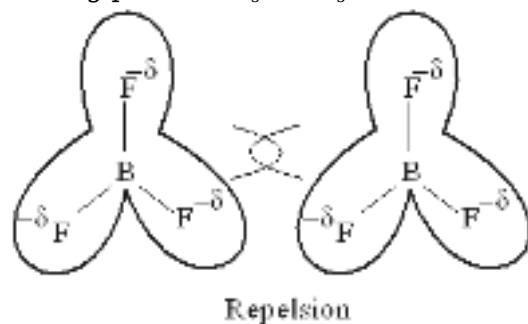
63) Inert gas + H₂O $\Rightarrow$ dipole-induced dipole

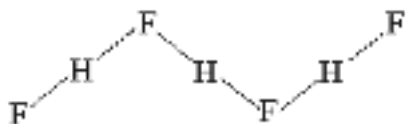
(Debye attraction)

64) (i), (ii) & (iii) are correct.

65)

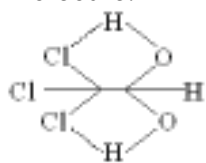
Boiling point BMe₃ > BF₃ $\rightarrow$ Deformation decrease the boiling point





Zig-zag due to H-bonding

The slippery nature of H_3BO_3 is due to layered structure formed by H-bonds between its molecule.



Chloral hydrate

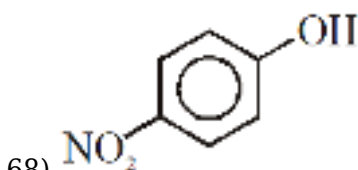
66)

Boiling point $\propto$ Intermolecular attraction forces

67) (A) Covalent character $\propto$ charge on cation

(B) Melting point $\propto$ Lattice energy

(C) Lattice energy $\propto \frac{\text{Charge on cation/anion}}{\text{radius on cation/anion}}$



68) has intermolecular H-bonds.

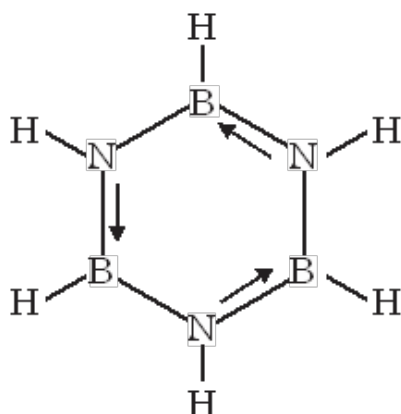
69) In Al_2Cl_6 hybridisation of Al is sp^3

70)

(a) $\text{N} \rightarrow \text{Si}$ ($2\text{p}\pi - 3\text{d}\pi$ back bond)

(b) $\text{P}(\text{SiH}_3)_3$ & SCl_2 -No back bonding

(d) OCl_2^- $2\text{p}\pi - 3\text{d}\pi$ back bonding



71)

Borazine

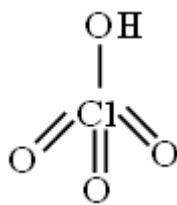
Inorganic Benzen

- Planar
- Aromatic (feebly)
- Back bonding
- Kinetically more reactive than Benzene but thermodynamically more stable than Benzene.

72) Octet of B is highly satisfied through strong back bonding by F where as octet of B is not fulfilled in case of BI_3 as I does weak back bond.

Hence BI_3 will accept easily lone pair hence will act strong lewis acid wheer as BF_3 will be weak lewis acid.

73) Fact



74) No peroxy linkage present.

75) $\text{NCl}_3 + \text{H}_2\text{O} \rightarrow \text{NH}_3 + \text{HOCl}$

$\text{BCl}_3 + \text{H}_2\text{O} \rightarrow \text{B(OH)}_3 + \text{HCl}$

76) **Explanation:** The question asks which of the following statements are correct about the stability and properties of certain compounds and ions.

Concept : (Stability order BCl_5 & BF_5 , PbCl_2 & GeCl_2 , SiO_2 & PbO_2 , & SnO_2 , Ti^{3+} ion & Ti^{4+} ion)

1. BiCl_5 is more stable than BiF_5 : INCORRECT. BiF_5 is known and relatively stable, while BiCl_5 is not stable. The +5 oxidation state of bismuth is more readily achieved with the highly electronegative and small fluorine atom. Chlorine is not able to stabilize the higher oxidation state due to the inert pair effect.

2. PbCl_2 , is more stable than GeCl_2 and SnCl_2 : CORRECT. Due to the inert pair effect, the +2 oxidation state becomes increasingly stable as you go down Group 14. Therefore, PbCl_2 is indeed more stable than GeCl_2 and SnCl_2 , which prefer the +4 state.

3. SiO_2 , is more acidic oxide than PbO_2 and SnO_2 : CORRECT. Across a period, acidity increases. Silicon is higher up in the group and further to the right than both tin and lead. So SiO_2 , is significantly more acidic. Both SnO_2 and PbO_2 , are amphoteric.

4. Tl^{3+} ion is more stable than Th^{+1} ion : INCORRECT. Thallium exhibits a strong preference for the +1 oxidation state due to the inert pair effect. Tl^{+1} is much more stable than Tl^{3+} .

77) Halide of 13th group act as Lewis acid

$\text{BCl}_3 > \text{AlCl}_3 > \text{GaCl}_3 > \text{InCl}_3$ - Lewis acidic strength

78) hybridisation of Al in $[\text{Al}(\text{H}_2\text{O})_6]^{+3}$ is sp^3d^2 .

79)

A is nonmetal while B is metal

80)

For alkaline earth metal e gain enthalpy is positive not negative

81)

Value of electron affinity is always less than ionisation energy

82)

Across a period non metallic character increases due to increasing electronegativity

83)

Process i, ii and iii are exothermic in nature while process iv is endothermic in nature

84)

e gain enthalpy of nitrogen is positive due to small size of 2p orbital in which incoming electron feels more repulsion while in P due to large size of 3p orbital incoming e feels less repulsion due to which for P e gain enthalpy is - ve.

85)

in halogen family EA order is $\text{Cl} > \text{F} > \text{Br} > \text{I}$

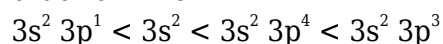
86)

Order of IE is $\text{F} > \text{Cl} > \text{Br} > \text{I}$

So, energy which ionises Cl also can ionise Br and I

87)

order of IE is



88)

Element having IUPAC name unnilpentium has official IUPAC name is Dubnium (Db)

89)

Given element belongs to f-block

90)

Apply magic number of group 10

BIOLOGY

91)

NCERT XI Page No. # 195

92) The correct answer is Option 2: 1-d-iii, 2-b-i, 3-a-iv, 4-c-ii

1. First heart sound (d LUB): - LUB is the sound associated with the first heart sound. - It occurs during ventricular systole (contraction) as the atrioventricular valves (mitral and tricuspid) close.

2. Cardiac output (b 5L): - Cardiac output is the volume of blood pumped by each ventricle per minute. - A typical value for cardiac output is around 5 liters per minute.

3. Second heart sound (a DUB): - DUB is the sound associated with the second heart sound. - It occurs during ventricular diastole (relaxation) as the semilunar valves (aortic and pulmonary) close.

4. Stroke volume (c 70 ml): - Stroke volume is the volume of blood pumped by each ventricle in one contraction. - A typical value for stroke volume is around 70 milliliters.

93) NCERT XI Pg # 195

94) NCERT, Pg # 196

95) NCERT Pg. # 198-199

96) NCERT Pg. No. # 200-201

97)

Pg. No. 196 XI-NCERT

98) **Question Explain :** Number of correct statements about heart and circulation.

Concept : This Question is based on Types Of Blood Circulation And Types Of Heart, Structure Of Human Heart

Explanation :

(A) Correct: Systemic circulation is from left ventricle to right atrium.

(B) Incorrect: Adrenal medullary hormones (like adrenaline) increase cardiac output.

(C) Correct: Heart failure is different from cardiac arrest or heart attack.

(D) Incorrect: Valve between right atrium and right ventricle is the tricuspid valve with three flaps, not two muscular flaps.

Final answer - 1. Two

99)

NCERT XI, Pg. No. 195

100)

NCERT Page No. 199

101) NCERT XI Page No. # 195

102) NCERT XI Page No. # 198

103) NCERT Pg # 195

104) NCERT Pg # 195

105) NCERT-XI, Pg # 194

106) NCERT Pg # 104

107) NCERT-XI, Pg. # 194

108) NCERT Pg.# 194

109) NCERT XIth Pg. # 194

110) NCERT (XI) Pg. # 194

111) NCERT (XI) Pg. # 187

112) NCERT (XI) Pg. # 190

113) NCERT XI, Page # 183

114)

NCERT XI, Pg. # 190

115)

NCERT XI Page # 185

116) NCERT (XI) Pg. # 268,269

117) NCERT (XI) Pg. # 273

118) NCERT (XI) Pg. # 274

119) NCERT (XIth) Pg. # 275, Para-4

120) A. Question: Identify the correct relationships between respiratory volumes and capacities.

B. Given Data:

■ (i) Inspiratory capacity (IC) = Tidal volume (TV) + Residual volume (RV)

■ (ii) Vital capacity (VC) = Tidal volume (TV) + Inspiratory reserve volume (IRV) + Expiratory reserve volume (ERV)

■ (iii) Residual volume (RV) = Vital Capacity (VC) - Inspiratory reserve volume (IRV)

■ (iv) Tidal volume (TV) = Inspiratory capacity (IC) - Inspiratory reserve volume (IRV)

C. Concept: Respiratory volumes and capacities.

D. Explanation:

(i) Inspiratory capacity (IC) = Tidal volume (TV) + Inspiratory reserve volume (IRV):

Inspiratory capacity is the maximum volume of air a person can inhale after a normal exhalation. It includes the tidal volume (the amount of air inhaled or exhaled during normal breathing) and the inspiratory reserve volume (the extra volume of air that can be inhaled beyond the tidal volume). The given equation incorrectly includes Residual volume (RV). So, statement (i) is incorrect.

(ii) Vital capacity (VC) = Tidal volume (TV) + Inspiratory reserve volume (IRV) + Expiratory reserve volume (ERV): Vital capacity is the maximum volume of air a person can exhale after a maximum inhalation. It includes the tidal volume, the inspiratory reserve volume, and the expiratory reserve volume (the extra volume of air that can be exhaled beyond the tidal volume). So, statement (ii) is correct.

(iii) Residual volume (RV) = Functional Residual Capacity (FRC) - Expiratory Reserve Volume (ERV): Residual volume is the volume of air remaining in the lungs after a maximal exhalation. It cannot be directly calculated from vital capacity. It is related to the Functional Residual Capacity (FRC) by the formula: $FRC = RV + ERV$. So, statement (iii) is incorrect.

(iv) Tidal volume (TV) = Inspiratory capacity (IC) - Inspiratory reserve volume (IRV): As explained in (i), $IC = TV + IRV$. Therefore, $TV = IC - IRV$. So, statement (iv) is correct.

121) NCERT Pg. # 268, Para 2

122)

NCERT XI Page # 270

123) New NCERT update

124)

125) **A. Question:** Which muscle contraction is responsible for normal expiration?

B. Given Data: Four muscles are listed as options.

C. Concept: This question tests knowledge of the mechanics of breathing, specifically expiration.

D. Explanation:

(A) Diaphragm: The diaphragm is the primary muscle of inspiration. During inspiration, it contracts and flattens, increasing the volume of the thoracic cavity. During normal expiration, the diaphragm relaxes and returns to its dome-shaped position.

(B) EICM (External Intercostal Muscles): EICMs are involved in inspiration, helping to expand the rib cage. They are not involved in normal expiration.

(C) IICM (Internal Intercostal Muscles): IICMs are involved in forceful expiration. During normal expiration, they are relaxed.

(D) Abdominal muscles: Abdominal muscles are used during forceful expiration, not normal expiration.

During normal expiration, no muscle contraction is required. It's a passive process that occurs due to the elastic recoil of the lungs and chest wall. The diaphragm simply relaxes, and the lungs deflate.

E. Final Answer: The correct answer is 4: No muscles contract during normal expiration

126) NCERT-XI , Pg# 275

127) A. Question: P_{50} value increases when:

B. Given Data: Options: 1. CO_2 conc. increase, 2. O_2 conc. increase, 3. CO_2 conc. decreases, 4. Temperature decrease

C. Concept: Oxygen-hemoglobin dissociation curve, P_{50} , and the Bohr effect.

D. Explanation: P_{50} is the partial pressure of oxygen at which 50% of hemoglobin is saturated with oxygen. It's a measure of the affinity of hemoglobin for oxygen.

Increased P_{50} : A rightward shift of the oxygen-hemoglobin dissociation curve indicates an increased P_{50} value. This means that a higher partial pressure of oxygen is required to achieve 50% hemoglobin saturation. This implies a lower affinity of hemoglobin for oxygen, favoring oxygen release to the tissues.

Factors that cause a rightward shift (increased P_{50}) and thus decrease hemoglobin's affinity for O_2 include:

- Increased CO_2 concentration (Bohr effect)
- Increased H^+ concentration (lower pH)
- Increased temperature
- Increased 2,3-DPG (diphosphoglycerate)

Conversely, a leftward shift (decreased P_{50}) means a lower partial pressure of oxygen is needed for 50% saturation, indicating higher affinity for oxygen.

E. Final Answer: 1. CO_2 conc. increase

128) NCERT-(XIth) Pg. # 269

129) **Explain Question :**

To identify the % of O_2 transported by blood plasma ,besides RBC

Concept : This question is based on transport of O₂

Explanation : About 97% of oxygen is transported bound to haemoglobin in RBCs, and the remaining ~3% is transported dissolved in plasma.

Final answer : 3

130) NCERT (XI) Pg # 286 (E)

131)

NCERT XI, Pg # 199, 200

132)

Concept :

pH - 7.4 (Slightly alkaline)

Solution/Explanation :

Due to presence of Bicarbonate ions in plasma.

133)

NCERT XI Pg#283

134)

NCERT XI, Pg # 287

135)

NCERT XI Pg. # 201

136)

Nucleolus, Nuclear pore, Nuclear membrane.

137)

NCERT Pg. No. 99

138) NCERT (XI) Pg. # 126, Para-2

139) NCERT-XI, Pg. # 132, Para-4, Line-1, 2, 7, 8, 9

140)

Explain Question : Which of the options represent the correct function of these organelles?

Concept : This question is based on endomembrane system.

Explanation : g. Aerobic Respiration → Mitochondria (not endomembrane)

h. Photosynthesis → Chloroplasts (not endomembrane)

Final answer : 1

141) NCERT Pg. # 97

142)

NCERT-XIth, Pg. # 96

143) NCERT, Pg. # 96, 97

144)

NCERT XI Pg. No. # 93 & 94

145) NCERT Page No. 164, 165, 10.2.1

146) NCERT-Pg. No. 169, 170

147) NCERT XI Eng. Pg # 162, 164, 166, 167, 169, 170

Hindi Pg # 162, 164, 166, 167, 169, 170

148) **Explanation**

A (Late prophase): (b) ER, Golgi complex disappear - During late prophase of meiosis, the endoplasmic reticulum (ER) and Golgi complex break down.

B (Anaphase): (a) Splitting of centromeres occurs - During anaphase centromeres split, allowing homologous chromosomes to separate.

C (Telophase): (d) Chromosomes decondense and lose their individuality - In telophase I, the chromosomes begin to decondense, and their distinct structure is lost.

D (Prophase): (c)

Longest phase of meiosis I - Prophase I is the longest phase of meiosis and involves complex events like synapsis and crossing over.

Correct answer: 2. A-b-i, B-a-iv, C-d-ii, D-c-iii

149)

Generated by Allie

Problem Statement: The question asks to identify the types of glycosidic bonds labeled as A and B in a diagram showing a portion of glycogen. It wants to know which bond corresponds to the 1-4 alpha and which corresponds to the 1-6 alpha glycosidic linkage.

Underlying Concept: Glycogen is a polysaccharide made up of glucose units linked primarily by alpha-1,4 glycosidic bonds forming the main chain. Branch points are created by alpha-1,6 glycosidic bonds. Understanding these types of linkages helps in comprehending glycogen's branched structure and how glucose units are connected.

Tips and Tricks: Remember that alpha-1,4 glycosidic bonds connect glucose units in straight chains, while alpha-1,6 glycosidic bonds are points of branching in glycogen.

Common Mistakes: Confusing the 1-4 bond with the 1-6 bond, or mixing up the position numbers in the glycosidic bond notation.

Why Other Options Are Incorrect?: Options that label A as the 1-6 bond and B as the 1-4 bond reverse the actual structure of glycogen, where linear linkages are 1-4 and branches are 1-6.

150)

NCERT-XIth, Pg. # 109, 110, 111

151)

NCERT Pg. # 143,147,148

152)

NCERT XI, Pg.# 104, 108, 109

153)

The correct answer is: Option 1 (Threonine, Valine, Lysine, and Tryptophan)

Explanation: Essential amino acids are those that cannot be synthesized by the body and must be obtained through the diet.

- A. **Threonine:** Essential
- B. **Valine:** Essential
- C. **Lysine:** Essential
- D. **Tryptophan:** Essential

154)

The correct answer is: Option 3 (III only)

Explanation:

Lactose: Lactose is made up of glucose and galactose linked by a **β(1-4) linkage**, and it is commonly found in **milk**.

155) NCERT-XI, Pg. # 115

156) NCERT Pg. # 114-117

157)

NCERT Pg. # 116

158) Question Explanation

Correct statement for polysaccharides.

Concept:

Polysaccharides are made up of monomer units Acid insoluble.

Soluton:

→ **Polysaccharides are acid insoluble.**

→ **These are long chains of monosaccharides**

→ **Cellulose is polysaccharides of β - D -Glucose (Monosaccharide unit)**

→ **Starch is a polymer and storage material (food) in plants.**

Correct option: (2)

159) NCERT Pg. # 126

160) NCERT Pg. # 125

161)

Meiocyte is a diploid structure and at G_1 -stage it contain $2C = 40$ Pg of DNA. The product of meiosis-II contain C amount of DNA. Hence, it will contain 20 pg of DNA.

162) NCERT Pg. # 126

163) NCERT pg. # 124

164)

Female gamete is haploid (n)

☐ **$n = 4$ chromosome**

As meiosis occur in diploid mother cell therefore it will contain 8 chromosome (or 4 bivalent)

165) NCERT pg. # 125 to 127

166) NCERT Pg. # 124

167) NCERT Pg. # 123

168)

NCERT-XIth, Pg. # 94,95,96,98

169) NCERT pg. # 98

170) NCERT Pg. # 112, 113

171) NCERT Pg. # 114, 115

172) NCERT Pg. # 117

173) NCERT Pg. # 118

174)

NCERT-XIth, Pg. # 110

175) Triose contain 3 carbon

Glucose contain 6 carbon

oligo saccharide posses 2 to 10 monomer starch is a polysaccharide

176) NCERT Pg. # 106

177) NCERT Pg. # 110

178) NCERT XI Pg. # 112

179) NCERT XI Pg. 126

180) Pollen grain is haploid (n) Hence, $n = 8$ chromosome & $C = 10$ pg DNA There force, microspore mother cell is diploid and contain $2n$ and $2c$